

Assignment1: CancerMortality

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Load the dataset and required libraries

```
rm(list=ls())
knitr::opts_chunk$set(fig.align='center', out.width='50%') # Manage output plots width and align

train <- read.csv("train.csv")
test  <- read.csv("test.csv")

library(moments)
library(car)

## Loading required package: carData
library(chemometrics)

## Loading required package: rpart
library(mice)

##
## Attaching package: 'mice'
## The following object is masked from 'package:stats':
##
##     filter
## The following objects are masked from 'package:base':
##
##     cbind, rbind
library(missMDA)
library(FactoMineR)
library(corrplot)

## corrplot 0.95 loaded
library(VIM)

## Loading required package: colorspace
## Loading required package: grid
## VIM is ready to use.
## Suggestions and bug-reports can be submitted at: https://github.com/statistikat/VIM/issues
##
## Attaching package: 'VIM'
## The following object is masked from 'package:datasets':
##
##     sleep
```

Define some functions to avoid repetition of same processes

```
calculate_outliers <- function(data) {
  # Check for regular outliers
  boxplot(variable, main = "Boxplot")
  outliers <- boxplot.stats(data)$out
}
```

```

# Calculate the lower and upper bounds for severe outliers
Q1 <- quantile(data, 0.25, na.rm = TRUE)
Q3 <- quantile(data, 0.75, na.rm = TRUE)
IQR <- Q3 - Q1

lower_severe_bound <- Q1 - 3 * IQR
upper_severe_bound <- Q3 + 3 * IQR
lower_mild_bound <- Q1 - 1.5 * IQR
upper_mild_bound <- Q3 + 1.5 * IQR

# Identify outliers
mild_outliers <- data[data < lower_mild_bound | data > upper_mild_bound]
severe_outliers <- data[data < lower_severe_bound | data > upper_severe_bound]
num_mild_outliers <- length(mild_outliers)
num_severe_outliers <- length(severe_outliers)

l11 <- which(data < lower_mild_bound | data > upper_mild_bound)

# Return the results as a list
return(list(
  num_outliers = num_mild_outliers,
  num_severe_outliers = num_severe_outliers,
  outliers_indexes = l11
))
}

summary_and_normality_check <- function(data) {
  #Basic summary
  print("Summary: ")
  print(summary(data))

  # Plot normality
  print("Mean")
  mm <- mean(data, na.rm = TRUE); print(mm)
  print("Standard deviation")
  ss <- sd(data, na.rm = TRUE); print(ss)
  hist(data,freq = F,100, main = "Histogram")
  curve(dnorm(x, mean = mm, sd = ss),col="red", lwd = 2, add=T)

  #Compute skewness
  print("Skewness")
  print(skewness(data, na.rm = TRUE))

  #Number of outliers
  out.res <- calculate_outliers(data)
  print("Num Outliers:")
  print(out.res$num_outliers)
  print("Num. severe outliers:")
  print(out.res$num_severe_outliers)

  #Shapiro test
  print("Shapiro test: ")
  print(shapiro.test(data))
}

```

```
    return(out.res$outliers_indexes)
  }

outliers_df <- train
outliers_df[,] <- 0
```

Data Preparation & Analysis

In this section, a descriptive analysis is conducted on each variable, a data quality report is generated, and imputation and profiling are addressed.

Univariate Descriptive Analysis (Variable Analysis) & Data quality

- Original numeric variables corresponding to qualitative concepts have to be converted to factors:
- Original numeric variables corresponding to real quantitative concepts are kept as numeric but additional factors should also be created
- Exploratory Data Analysis for each variable (numeric summary and graphic support) of the train sample.

Variable *avganncount*

avganncount is a numerical variable representing the “Mean number of reported cancer cases diagnosed annually (2010-2015).” The difference between the mean (623.2) and the median (175) indicates that the distribution is not symmetric and deviates from normality. Additionally, the first and third quartiles (80 and 509, respectively) are much lower than the mean of 623.2 and the maximum value of 38150.0, further suggesting skewness. The boxplot confirms this skewness, with an extended upper whisker and numerous outliers, indicating a heavy right tail. It has 333 outliers, and 257 of these are severe outliers.

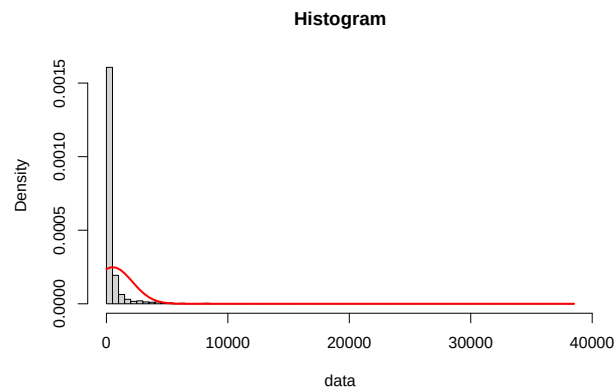
In this variable there are 126 values = “1962.667684” that considered as NA . Lastly, the skeweness indicator and Shapiro test rejection of the null hypothesis also confirm the non-normal distribution of the variable.

```
variable <- train$avganncount
l11 <- which(variable == 1962.667684 ); length(l11)
```

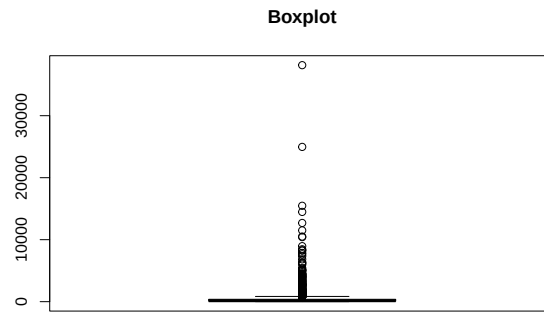
```
## [1] 126
```

```
variable[variable == 1962.667684 ] <- NA
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##      7.0   74.0   155.0   524.2   383.0 38150.0    126
## [1] "Mean"
## [1] 524.2393
## [1] "Standard deviation"
## [1] 1601.779
```



```
## [1] "Skewness"
## [1] 12.16498
```



```
## [1] "Num Outliers:"
## [1] 333
## [1] "Num. severe outliers:"
## [1] 257
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.2677, p-value < 2.2e-16
```

```
outliers_df[res.var,"avganncount"] <- 1
outliers <- which(outliers_df$avganncount == 1); length(outliers)
```

```
## [1] 207
```

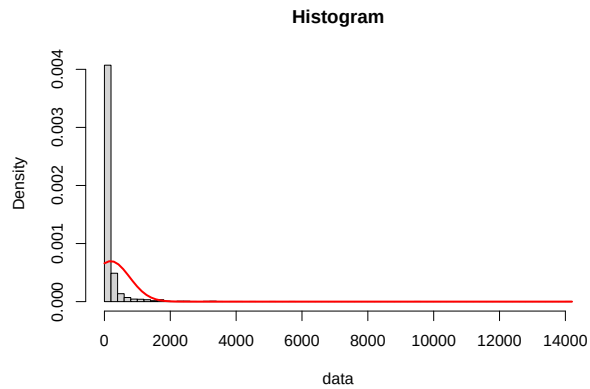
Variable *avgdeathsperyear*

In a similar way, *avgdeathsperyear* is a numerical variable expressing the “Mean number of reported mortalities due to cancer”.

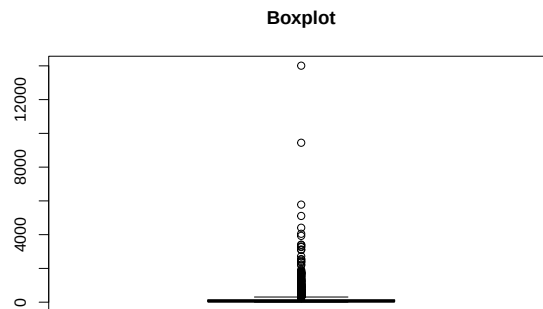
It has no missing values. The difference between the mean (191.6) and median (62) indicates that the distribution is not symmetric. Moreover, the first and third quartiles (29 and 140.5, respectively) are lower than the mean and the maximum value of 14010, further suggesting a skewed distribution. With the help of graphical tools like the boxplot we can confirm the skewness, obtaining 225 outliers (138 of which are severe), indicating a large right tail. The histogram overlaid with a normal curve highlights the deviation from a normal distribution with the same mean and standard deviation, and the Shapiro test also confirms the non-normality of the variable.

```
variable <- train$avgdeathsperyear;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   3.0   29.0   62.0  191.6  140.5 14010.0
## [1] "Mean"
## [1] 191.6237
## [1] "Standard deviation"
## [1] 571.8763
```



```
## [1] "Skewness"
## [1] 12.4999
```



```
## [1] "Num Outliers:"
## [1] 225
## [1] "Num. severe outliers:"
## [1] 138
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.26769, p-value < 2.2e-16
```

```
outliers_df[res.var,"avgdeathsperyear"] <- 1
```

Variable *target_deathrate* :

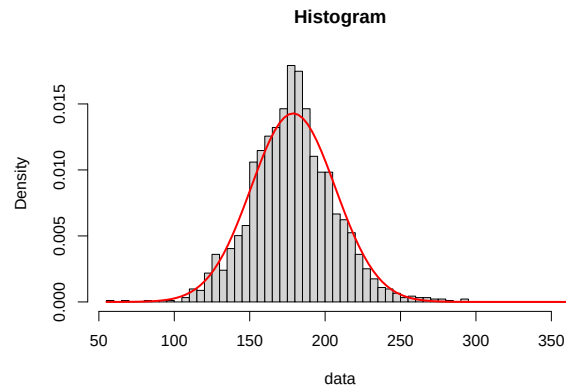
Variable *target_deathrate* is a numerical variable that is the target of the model. It contains no missing values in the train sample. The median (178.3) and mean (178.8) values are really close, expressing the symmetry of the distribution of the variable. Thanks to the boxplot and the histogram we can graphically confirm the symmetry of the distribution.

We performed the Shapiro test to check for the normality of the data, and obtained a small p-value that rejects the null hypothesis, indicating that the variable does not follow a normal distribution. Therefore, the target variable is not normally distributed.

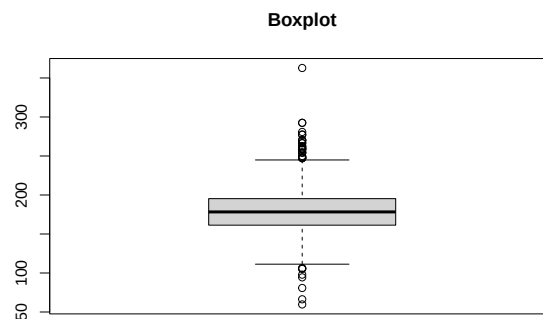
Moreover, it contains 35 outliers, 1 of which is a severe.

```
variable <- train$target_deathrate;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      59.7  161.3   178.3   178.8   195.3   362.8
## [1] "Mean"
## [1] 178.8197
## [1] "Standard deviation"
## [1] 27.93042
```



```
## [1] "Skewness"
## [1] 0.3243216
```



```
## [1] "Num Outliers:"
## [1] 35
## [1] "Num. severe outliers:"
## [1] 1
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98647, p-value = 4.149e-12
```

```
outliers_df[res.var,"target_deathrate"] <- 1
```

Variable *incidencerate* :

Variable *incidencerate* is a numerical variable that expresses “Mean per capita (100,000) cancer diagnoses”.

The variable has no missing values. It has similar mean (449) and median (453.5) values given its range and suggesting a symmetry distribution. It contains 60 outliers, 7 of which are severe outliers. In the boxplot we can see how the most severe outliers are high and not low values, corresponding to Osceola County, Florida and Williamsburg city, Virginia. The histogram with the overlapping normal distribution shows a quite symmetrical distribution, but the Shapiro test and skeweness value discard the normality of the variable's distribution.

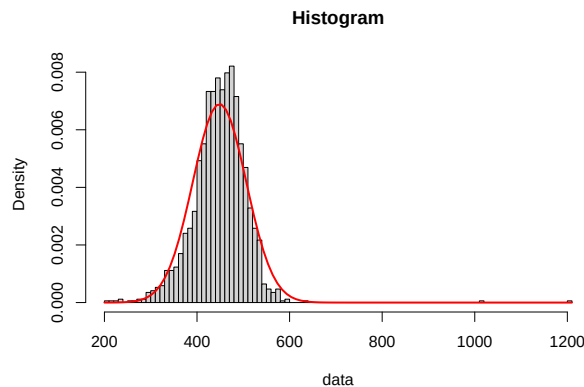
In this variable there are 126 values = “453.5494221” we should consider these as NA (same rows as avganncount variable)

```
variable <- train$incidencerate;  
l11 <- which(variable == 453.5494221 ); length(l11)
```

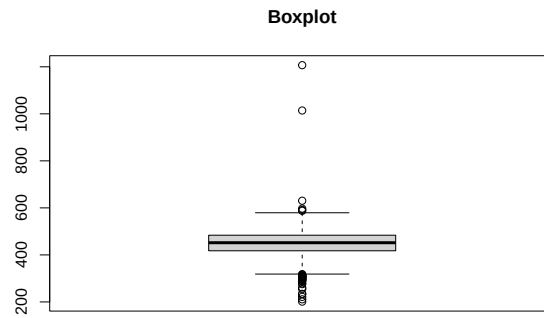
```
## [1] 126
```

```
variable[variable == 453.5494221 ] <- NA  
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's  
##      201.3  417.5   451.8   448.7  483.8  1206.9     126  
## [1] "Mean"  
## [1] 448.6551  
## [1] "Standard deviation"  
## [1] 57.85323
```



```
## [1] "Skewness"  
## [1] 1.397679
```



```
## [1] "Num Outliers:"
## [1] 162
## [1] "Num. severe outliers:"
## [1] 130
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.90204, p-value < 2.2e-16
outliers_df[res.var,"incidencerate"] <- 1
```

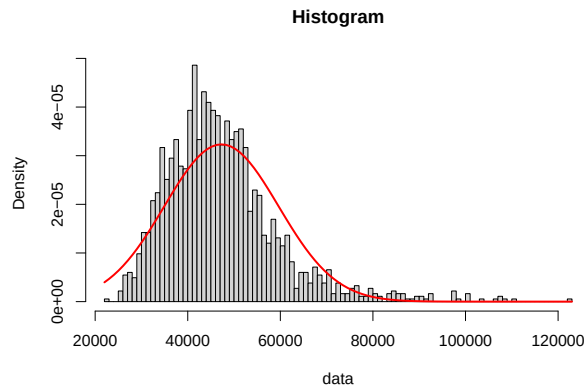
Variable *medincome* :

Variable *medincome* is a numerical variable that expresses the “Median income per county”.

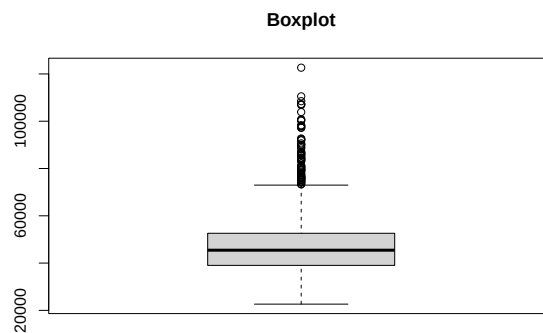
Thanks to the histogram, Shapiro test, and skeweness value, we see how the distribution is neither Normal nor symmetric, but it is skewed with a long right tail. It has 69 outliers, 15 of them severe. The highest value is for Loudoun County, Virginia. There are no missing values in this variable.

```
variable <- train$medincome;
res.var <- summary_and_normality_check(variable)

## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 22640  39031  45454   47278  52612 122641
## [1] "Mean"
## [1] 47277.94
## [1] "Standard deviation"
## [1] 12341.69
```



```
## [1] "Skewness"
## [1] 1.466811
```



```
## [1] "Num Outliers:"
## [1] 69
## [1] "Num. severe outliers:"
## [1] 15
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.9105, p-value < 2.2e-16
outliers_df[res.var,"medincome"] <- 1
```

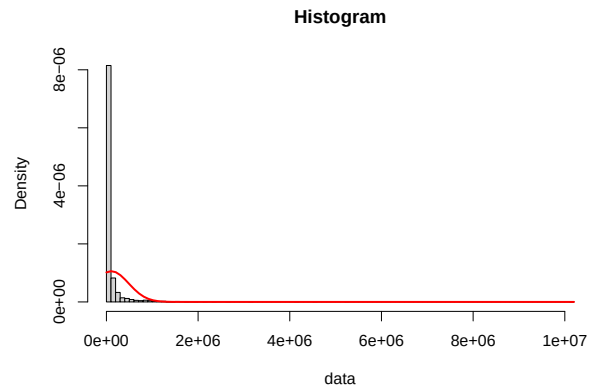
Variable *popest2015* :

Variable *popest2015* is a numerical variable that expresses the population of the county.

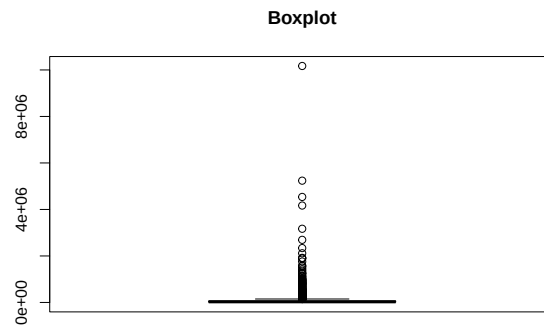
It does not count with missing values. It has a long range and is highly skewed, with a long right tail. In fact, it contains 252 outliers, 162 of which are severe, and all of them are above the upper whisker in the boxplot. The highest value is for Los Angeles County, California. The large difference between the mean (106.841) and median (27.158), along with the spread between the 1st (12.191) and 3rd quartiles (66.880), indicates a positively skewed distribution, where extreme high values are pulling the mean upward.

```
variable <- train$popest2015;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      829  12191   27158  106841  66880 10170292
## [1] "Mean"
## [1] 106841.2
## [1] "Standard deviation"
## [1] 376494
```



```
## [1] "Skewness"
## [1] 14.70818
```



```
## [1] "Num Outliers:"
## [1] 252
## [1] "Num. severe outliers:"
## [1] 162
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.22666, p-value < 2.2e-16
```

```
outliers_df[res.var,"popest2015"] <- 1
```

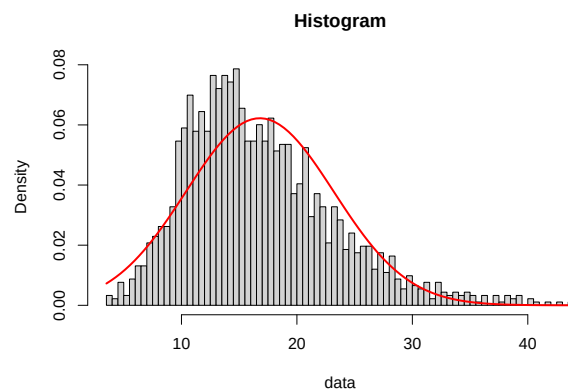
Variable *povertypersent* :

Variable *povertypersent* is a numerical variable that expresses the percent of population in poverty and has no missing values.

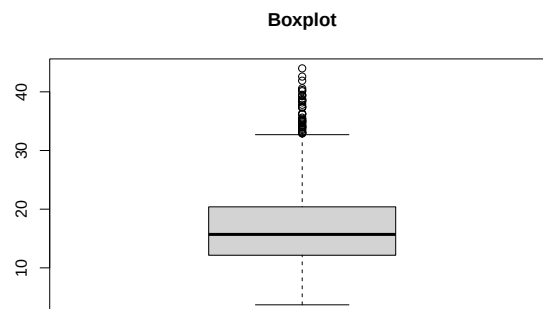
Similarly to other variables, it has a skewed distribution with a larger right tail. It has 42 outliers, but none of them is severe. The p-value obtained in the shapiro_ test confirms the rejection of the null hypothesis, concluding that the variable does not have a normal distribution.

```
variable <- train$povertypersent;  
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "  
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##   3.70  12.15   15.70   16.79  20.40   44.00  
## [1] "Mean"  
## [1] 16.79175  
## [1] "Standard deviation"  
## [1] 6.411161
```



```
## [1] "Skewness"  
## [1] 0.8905168
```



```
## [1] "Num Outliers:"  
## [1] 42  
## [1] "Num. severe outliers:"  
## [1] 0  
## [1] "Shapiro test: "
```

```
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.95557, p-value < 2.2e-16
outliers_df[res.var,"povertypercent"] <- 1
```

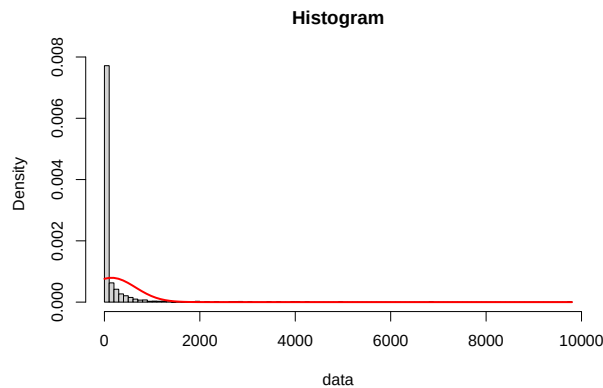
Variable *studypcap* : I CONSIDER IT IS WORTH TO REMOVE

Variable *studypcap* is a numerical variable that expresses “Per capita number of cancer-related clinical trials per county”.

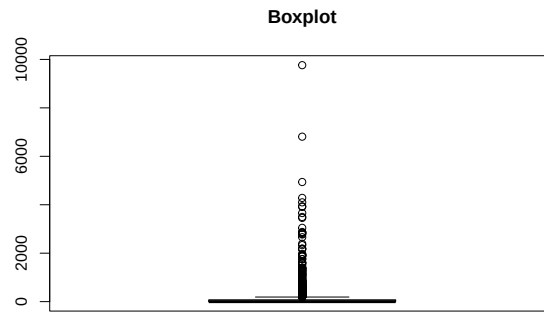
It contains no missing values, and has 307 outliers (222 severe). Its distribution is non-symmetric. Note that 1162 out of 1831 observations contain value 0 in this variable, which indicates that it may only provide noise to the model. **Thus, we consider it is worth to remove the variable.**

```
variable <- train$studypcap;
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.0     0.0     0.0   148.2   76.0  9762.3
## [1] "Mean"
## [1] 148.2024
## [1] "Standard deviation"
## [1] 501.867
```



```
## [1] "Skewness"
## [1] 8.477802
```



```
## [1] "Num Outliers:"
## [1] 307
## [1] "Num. severe outliers:"
## [1] 222
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.30754, p-value < 2.2e-16
outliers_df[res.var,"studypercap"] <- 1
```

Variable *binnedi* : I CONSIDER IT IS WORTH TO REMOVE

Variable *binnedi* is a categorical variable (or factor) that expresses the median income per capita binned by decile.

```
variable <- train$binnedinc;
summary <- summary(variable); summary
```

```
##      Length      Class      Mode
##      1831 character character
```

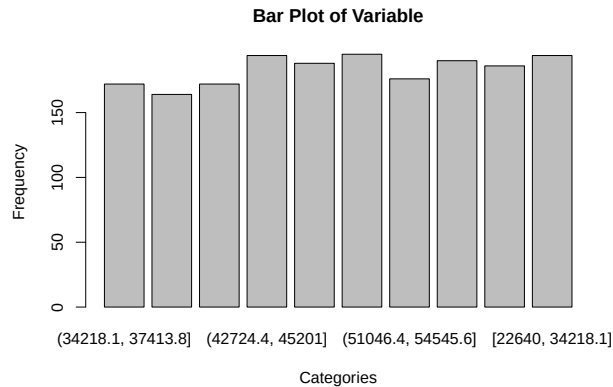
```
table(variable)
```

```
## variable
## (34218.1, 37413.8] (37413.8, 40362.7] (40362.7, 42724.4] (42724.4, 45201]
##              172              164              172              194
## (45201, 48021.6] (48021.6, 51046.4] (51046.4, 54545.6] (54545.6, 61494.5]
##              188              195              176              190
## (61494.5, 125635] [22640, 34218.1]
##              186              194
```

```
prop.table(table(variable))
```

```
## variable
## (34218.1, 37413.8] (37413.8, 40362.7] (40362.7, 42724.4] (42724.4, 45201]
##      0.09393774      0.08956854      0.09393774      0.10595303
## (45201, 48021.6] (48021.6, 51046.4] (51046.4, 54545.6] (54545.6, 61494.5]
##      0.10267613      0.10649918      0.09612234      0.10376843
## (61494.5, 125635] [22640, 34218.1]
##      0.10158383      0.10595303
```

```
barplot(table(variable), main="Bar Plot of Variable", xlab="Categories", ylab="Frequency")
```



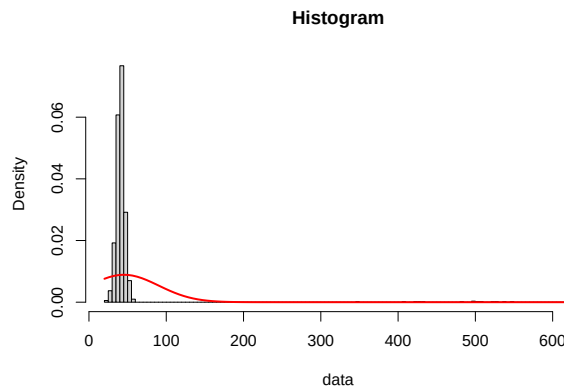
Variable *medianage* :

Medianage is a numerical variable that expresses the median age of county residents.

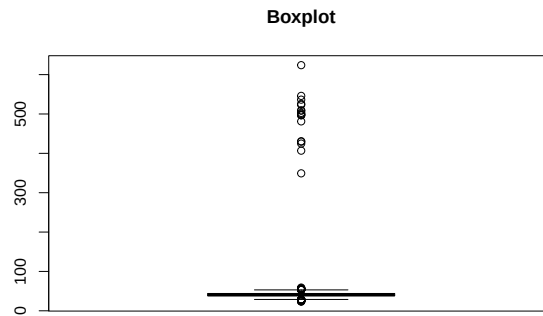
The variable has no missing values. Has 65 outliers, 18 of which are severe outliers. All of these outliers should be removed because it's impossible to be more than 100 years. Once we removed all the incorrect values we get 52 outliers, 18 of then severe. Finally we observe a symmetric distribution with a median = 40.90 and a mean=40.85. After analyzing the boxplot, and the histogram along the Shapiro test we can discard the normality of the variable's distribution.

```
variable <- train$medianage
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   23.30  37.90   40.90   45.25  44.00   624.00
## [1] "Mean"
## [1] 45.2497
## [1] "Standard deviation"
## [1] 44.85337
```



```
## [1] "Skewness"
## [1] 9.996575
```



```
## [1] "Num Outliers:"
## [1] 65
## [1] "Num. severe outliers:"
## [1] 18
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.14066, p-value < 2.2e-16
outliers_df[res.var,"medianage"] <- 1
```

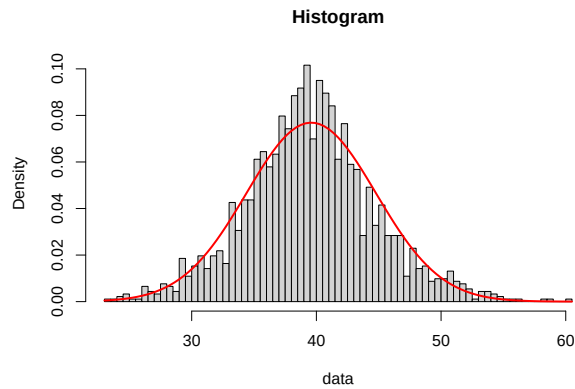
Variable *medianagemale* :

Numeric variable, represents the mean age of the male population of each county.

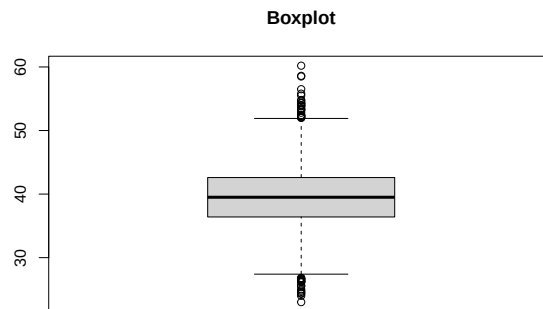
There are no missing values and it has 46 outliers, non of them several outliers. The mean is 39.59 yo and the median 39.50yo what suggest a simmetry distribution. After looking at the results of the shapiro-test it doesn't have a normal distribution

```
variable <- train$medianagemale
res.var <- summary_and_normality_check(variable)

## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   23.00  36.40   39.50   39.59  42.60   60.20
## [1] "Mean"
## [1] 39.58564
## [1] "Standard deviation"
## [1] 5.188667
```



```
## [1] "Skewness"
## [1] 0.1612965
```



```
## [1] "Num Outliers:"
## [1] 46
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99404, p-value = 9.877e-07
```

```
outliers_df[res.var, "medianagemale"] <- 1
```

Variable *medianagemale* : I AM NOT SURE IF IT'S WORTH IT TO SEPARATE THE MEDIANAGE WITH SEX

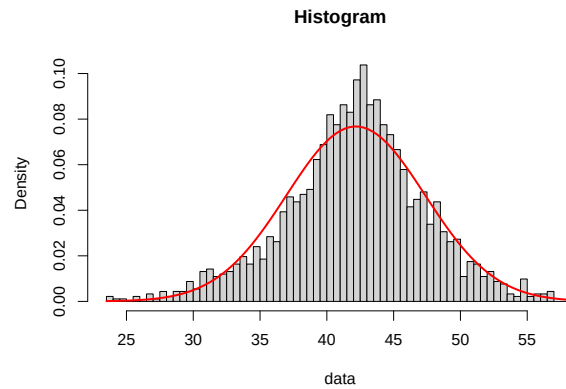
The *medianagemale* is a numerical variable that analyze the median age of the females in each county.

It has a simmetric distribution (median=42.40, mean=42,17) but after analyzing the results of the Shapiro-test we observe that does not have a normal distribution.

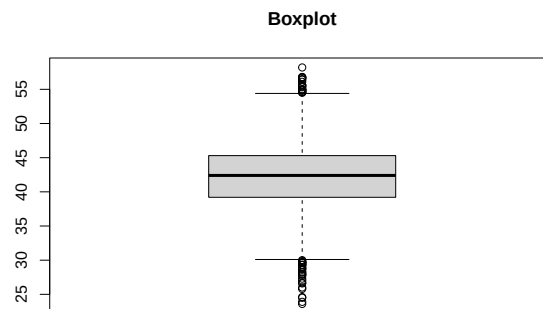
```
variable <- train$medianagemale
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    23.60  39.20   42.40   42.17  45.30   58.20
## [1] "Mean"
## [1] 42.16931
## [1] "Standard deviation"
## [1] 5.196007
```



```
## [1] "Skewness"
## [1] -0.2057034
```



```
## [1] "Num Outliers:"
## [1] 55
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
##  Shapiro-Wilk normality test
##
## data:  data
## W = 0.99321, p-value = 1.817e-07
```

```
outliers_df[res.var,"medianagefemale"] <- 1
```

Variable *geography* :

Categorical variable that shows the name of each county. We can use this variable as a PK

-comment - Maybe we can separate the name of the state and create a new variable to be able to study the distribution around the United States

```
variable <- train$geography;
summary <- summary(variable); summary
```

```
##      Length      Class      Mode
##      1831 character character
```

```
table(variable)
```

```
## variable
##          Acadia Parish, Louisiana          Ada County, Idaho
##                      1                      1
##          Adair County, Kentucky          Adair County, Missouri
##                      1                      1
##          Adams County, Idaho          Adams County, Illinois
##                      1                      1
##          Adams County, Iowa          Adams County, Mississippi
##                      1                      1
##          Adams County, Nebraska          Adams County, North Dakota
##                      1                      1
##          Adams County, Ohio          Aiken County, South Carolina
##                      1                      1
##          Alachua County, Florida          Albany County, Wyoming
##                      1                      1
##          Albemarle County, Virginia          Alcona County, Michigan
##                      1                      1
##          Alcorn County, Mississippi          Aleutians West Census Area, Alaska
##                      1                      1
##          Alexander County, Illinois          Alfalfa County, Oklahoma
##                      1                      1
##          Alger County, Michigan          Allegan County, Michigan
##                      1                      1
##          Allegany County, New York          Alleghany County, Virginia
##                      1                      1
##          Allen County, Kansas          Allen Parish, Louisiana
##                      1                      1
##          Alpena County, Michigan          Amador County, California
##                      1                      1
##          Amelia County, Virginia          Amite County, Mississippi
##                      1                      1
##          Anchorage Municipality, Alaska          Anderson County, Kentucky
##                      1                      1
##          Anderson County, South Carolina          Andrew County, Missouri
##                      1                      1
##          Androscoggin County, Maine          Anne Arundel County, Maryland
##                      1                      1
##          Anoka County, Minnesota          Anson County, North Carolina
##                      1                      1
##          Antelope County, Nebraska          Antrim County, Michigan
##                      1                      1
##          Appanoose County, Iowa          Appling County, Georgia
##                      1                      1
##          Arapahoe County, Colorado          Archer County, Texas
```

##	1	1
##	Archuleta County, Colorado	Arenac County, Michigan
##	1	1
##	Arlington County, Virginia	Armstrong County, Pennsylvania
##	1	1
##	Armstrong County, Texas	Aroostook County, Maine
##	1	1
##	Ascension Parish, Louisiana	Ashe County, North Carolina
##	1	1
##	Ashland County, Ohio	Ashtabula County, Ohio
##	1	1
##	Assumption Parish, Louisiana	Atchison County, Kansas
##	1	1
##	Atchison County, Missouri	Athens County, Ohio
##	1	1
##	Atlantic County, New Jersey	Atoka County, Oklahoma
##	1	1
##	Attala County, Mississippi	Audubon County, Iowa
##	1	1
##	Augusta County, Virginia	Aurora County, South Dakota
##	1	1
##	Austin County, Texas	Avoyelles Parish, Louisiana
##	1	1
##	Baca County, Colorado	Bacon County, Georgia
##	1	1
##	Baker County, Florida	Baker County, Georgia
##	1	1
##	Baker County, Oregon	Baltimore city, Maryland
##	1	1
##	Baltimore County, Maryland	Bamberg County, South Carolina
##	1	1
##	Banks County, Georgia	Bannock County, Idaho
##	1	1
##	Barbour County, Alabama	Barbour County, West Virginia
##	1	1
##	Barnes County, North Dakota	Barren County, Kentucky
##	1	1
##	Barron County, Wisconsin	Barrow County, Georgia
##	1	1
##	Barry County, Michigan	Barton County, Missouri
##	1	1
##	Bastrop County, Texas	Bates County, Missouri
##	1	1
##	Bath County, Virginia	Bayfield County, Wisconsin
##	1	1
##	Beadle County, South Dakota	Bear Lake County, Idaho
##	1	1
##	Beaufort County, North Carolina	Beaufort County, South Carolina
##	1	1
##	Beaver County, Oklahoma	Beaver County, Utah
##	1	1
##	Beaverhead County, Montana	Bedford County, Tennessee
##	1	1
##	Bee County, Texas	Belknap County, New Hampshire

##	1	1
##	Bell County, Texas	Beltrami County, Minnesota
##	1	1
##	Benewah County, Idaho	Bennington County, Vermont
##	1	1
##	Benson County, North Dakota	Benton County, Indiana
##	1	1
##	Benton County, Minnesota	Benton County, Mississippi
##	1	1
##	Benton County, Missouri	Benton County, Oregon
##	1	1
##	Benton County, Tennessee	Benton County, Washington
##	1	1
##	Berkeley County, South Carolina	Berkeley County, West Virginia
##	1	1
##	Berkshire County, Massachusetts	Bernalillo County, New Mexico
##	1	1
##	Berrien County, Georgia	Berrien County, Michigan
##	1	1
##	Bertie County, North Carolina	Bexar County, Texas
##	1	1
##	Bibb County, Georgia	Bienville Parish, Louisiana
##	1	1
##	Big Horn County, Montana	Big Horn County, Wyoming
##	1	1
##	Black Hawk County, Iowa	Blaine County, Idaho
##	1	1
##	Blaine County, Oklahoma	Blair County, Pennsylvania
##	1	1
##	Blanco County, Texas	Blue Earth County, Minnesota
##	1	1
##	Bolivar County, Mississippi	Bon Homme County, South Dakota
##	1	1
##	Bonner County, Idaho	Boone County, Arkansas
##	1	1
##	Boone County, Illinois	Boone County, Indiana
##	1	1
##	Bosque County, Texas	Botetourt County, Virginia
##	1	1
##	Bottineau County, North Dakota	Bourbon County, Kansas
##	1	1
##	Bowman County, North Dakota	Box Elder County, Utah
##	1	1
##	Boyd County, Kentucky	Boyle County, Kentucky
##	1	1
##	Bradford County, Pennsylvania	Bradley County, Arkansas
##	1	1
##	Bradley County, Tennessee	Brantley County, Georgia
##	1	1
##	Braxton County, West Virginia	Breathitt County, Kentucky
##	1	1
##	Breckinridge County, Kentucky	Bremer County, Iowa
##	1	1
##	Brevard County, Florida	Bristol County, Rhode Island

##	1	1
##	Broadwater County, Montana	Brookings County, South Dakota
##	1	1
##	Brooks County, Georgia	Brooks County, Texas
##	1	1
##	Broome County, New York	Broomfield County, Colorado
##	1	1
##	Broward County, Florida	Brown County, Kansas
##	1	1
##	Brown County, Ohio	Brown County, Texas
##	1	1
##	Brule County, South Dakota	Brunswick County, North Carolina
##	1	1
##	Brunswick County, Virginia	Bryan County, Oklahoma
##	1	1
##	Buchanan County, Iowa	Buchanan County, Missouri
##	1	1
##	Buchanan County, Virginia	Buckingham County, Virginia
##	1	1
##	Buena Vista County, Iowa	Buffalo County, Wisconsin
##	1	1
##	Bullitt County, Kentucky	Bulloch County, Georgia
##	1	1
##	Bullock County, Alabama	Buncombe County, North Carolina
##	1	1
##	Burke County, Georgia	Burke County, North Dakota
##	1	1
##	Burleson County, Texas	Burlington County, New Jersey
##	1	1
##	Burnet County, Texas	Burnett County, Wisconsin
##	1	1
##	Burt County, Nebraska	Butler County, Ohio
##	1	1
##	Butte County, South Dakota	Cabarrus County, North Carolina
##	1	1
##	Caddo County, Oklahoma	Calaveras County, California
##	1	1
##	Calcasieu Parish, Louisiana	Caldwell County, Kentucky
##	1	1
##	Caldwell County, Missouri	Caldwell County, North Carolina
##	1	1
##	Caldwell County, Texas	Caldwell Parish, Louisiana
##	1	1
##	Calhoun County, Arkansas	Calhoun County, Georgia
##	1	1
##	Calhoun County, Iowa	Calhoun County, Michigan
##	1	1
##	Calhoun County, South Carolina	Calhoun County, West Virginia
##	1	1
##	Callahan County, Texas	Calumet County, Wisconsin
##	1	1
##	Camden County, Missouri	Camden County, North Carolina
##	1	1
##	Cameron County, Pennsylvania	Cameron Parish, Louisiana

##	1	1
##	Campbell County, Kentucky	Campbell County, Virginia
##	1	1
##	Campbell County, Wyoming	Canadian County, Oklahoma
##	1	1
##	Cannon County, Tennessee	Cape Girardeau County, Missouri
##	1	1
##	Cape May County, New Jersey	Carbon County, Montana
##	1	1
##	Carbon County, Utah	Carbon County, Wyoming
##	1	1
##	Carlisle County, Kentucky	Carlton County, Minnesota
##	1	1
##	Caroline County, Virginia	Carroll County, Georgia
##	1	1
##	Carroll County, Illinois	Carroll County, Indiana
##	1	1
##	Carroll County, Iowa	Carroll County, Kentucky
##	1	1
##	Carroll County, Maryland	Carroll County, New Hampshire
##	1	1
##	Carroll County, Ohio	Carroll County, Tennessee
##	1	1
##	Carroll County, Virginia	Carson City, Nevada
##	1	1
##	Carson County, Texas	Carter County, Kentucky
##	1	1
##	Carter County, Tennessee	Carver County, Minnesota
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##	Casey County, Kentucky	Cass County, Iowa
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##	Cass County, Michigan	Cass County, Missouri
##	1	1
##	Cass County, Nebraska	Cass County, North Dakota
##	1	1
##	Cass County, Texas	Cassia County, Idaho
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##	Castro County, Texas	Caswell County, North Carolina
##	1	1
##	Catahoula Parish, Louisiana	Catawba County, North Carolina
##	1	1
##	Catoosa County, Georgia	Cattaraugus County, New York
##	1	1
##	Cavalier County, North Dakota	Cayuga County, New York
##	1	1
##	Cedar County, Iowa	Cedar County, Missouri
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##	Cedar County, Nebraska	Centre County, Pennsylvania
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##	Cerro Gordo County, Iowa	Chaffee County, Colorado
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##	Chambers County, Alabama	Chambers County, Texas
##	1	1
##	Champaign County, Ohio	Chariton County, Missouri

##	1	1
##	Charles Mix County, South Dakota	Charleston County, South Carolina
##	1	1
##	Charlotte County, Virginia	Charlton County, Georgia
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##	Chase County, Kansas	Chatham County, Georgia
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##	Chattahoochee County, Georgia	Chattooga County, Georgia
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##	Cheatham County, Tennessee	Cheboygan County, Michigan
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##	Chelan County, Washington	Chemung County, New York
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##	Chenango County, New York	Cherokee County, Iowa
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##	Cherokee County, North Carolina	Cherokee County, Oklahoma
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##	Cherokee County, South Carolina	Chester County, Pennsylvania
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##	Chester County, South Carolina	Chester County, Tennessee
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##	Chesterfield County, South Carolina	Cheyenne County, Colorado
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##	Chickasaw County, Iowa	Chickasaw County, Mississippi
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##	Childress County, Texas	Chilton County, Alabama
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##	Chippewa County, Michigan	Chippewa County, Minnesota
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##	Chisago County, Minnesota	Chittenden County, Vermont
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##	Choctaw County, Oklahoma	Chouteau County, Montana
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##	Churchill County, Nevada	Cibola County, New Mexico
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##	Cimarron County, Oklahoma	Citrus County, Florida
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##	Claiborne County, Mississippi	Claiborne County, Tennessee
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##	Clallam County, Washington	Clare County, Michigan
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##	Clarendon County, South Carolina	Clark County, Missouri
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##	Clark County, Nevada	Clark County, South Dakota
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##	Clark County, Washington	Clark County, Wisconsin
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##	Clarke County, Georgia	Clarke County, Mississippi
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##	Clay County, Florida	Clay County, Georgia
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##	Clay County, Illinois	Clay County, Iowa
##	1	1
##	Clay County, Kansas	Clay County, Kentucky

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##	Clay County, Minnesota	Clay County, Mississippi
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##	Clay County, Tennessee	Clay County, Texas
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##	Clay County, West Virginia	Clayton County, Georgia
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##	Clayton County, Iowa	Clearfield County, Pennsylvania
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##	Cleburne County, Alabama	Cleburne County, Arkansas
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##	Cleveland County, North Carolina	Clinch County, Georgia
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##	Clinton County, Iowa	Clinton County, Michigan
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##	Clinton County, New York	Clinton County, Pennsylvania
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##	Cloud County, Kansas	Coahoma County, Mississippi
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##	Cochise County, Arizona	Cochran County, Texas
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##	Cocke County, Tennessee	Coconino County, Arizona
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##	Coffee County, Alabama	Coffee County, Georgia
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##	Coffee County, Tennessee	Coffey County, Kansas
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##	Colbert County, Alabama	Colfax County, Nebraska
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##	Colfax County, New Mexico	Collier County, Florida
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##	Colonial Heights city, Virginia	Colquitt County, Georgia
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##	Columbia County, Georgia	Columbia County, New York
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##	Columbia County, Oregon	Columbia County, Washington
##	1	1
##	Columbia County, Wisconsin	Columbiana County, Ohio
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##	Columbus County, North Carolina	Colusa County, California
##	1	1
##	Comal County, Texas	Concho County, Texas
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##	Concordia Parish, Louisiana	Contra Costa County, California
##	1	1
##	Conway County, Arkansas	Cook County, Georgia
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##	Cook County, Illinois	Cook County, Minnesota
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##	Cooke County, Texas	Cooper County, Missouri
##	1	1
##	Coos County, New Hampshire	Coosa County, Alabama

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##	Corson County, South Dakota	Cortland County, New York
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##	Coryell County, Texas	Cottle County, Texas
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##	Covington County, Mississippi	Cowley County, Kansas
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##	Cowlitz County, Washington	Crane County, Texas
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##	Craven County, North Carolina	Crawford County, Arkansas
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##	Crawford County, Georgia	Crawford County, Indiana
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##	Crawford County, Iowa	Crawford County, Michigan
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##	Crawford County, Ohio	Crawford County, Wisconsin
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##	Crenshaw County, Alabama	Crisp County, Georgia
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##	Crittenden County, Arkansas	Crittenden County, Kentucky
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##	Crockett County, Tennessee	Crockett County, Texas
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##	Cross County, Arkansas	Crow Wing County, Minnesota
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##	Cullman County, Alabama	Culpeper County, Virginia
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##	Cumberland County, Kentucky	Cumberland County, Maine
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##	Cumberland County, Tennessee	Cuming County, Nebraska
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##	Curry County, New Mexico	Curry County, Oregon
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##	Custer County, Colorado	Custer County, Oklahoma
##	1	1
##	Cuyahoga County, Ohio	Dade County, Georgia
##	1	1
##	Dade County, Missouri	Dakota County, Minnesota
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##	Dale County, Alabama	Dallam County, Texas
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##	Dallas County, Alabama	Dallas County, Iowa
##	1	1
##	Dallas County, Missouri	Dane County, Wisconsin
##	1	1
##	Danville city, Virginia	Dare County, North Carolina
##	1	1
##	Darlington County, South Carolina	Davidson County, North Carolina
##	1	1
##	Daviess County, Indiana	Daviess County, Kentucky
##	1	1
##	Daviess County, Missouri	Davis County, Utah
##	1	1
##	Davison County, South Dakota	Dawes County, Nebraska

##	1	1
##	Dawson County, Montana	Dawson County, Nebraska
##	1	1
##	Dawson County, Texas	Day County, South Dakota
##	1	1
##	De Baca County, New Mexico	De Soto Parish, Louisiana
##	1	1
##	Deaf Smith County, Texas	Dearborn County, Indiana
##	1	1
##	Decatur County, Indiana	Decatur County, Iowa
##	1	1
##	Decatur County, Kansas	Decatur County, Tennessee
##	1	1
##	Defiance County, Ohio	DeKalb County, Alabama
##	1	1
##	DeKalb County, Georgia	DeKalb County, Illinois
##	1	1
##	DeKalb County, Indiana	DeKalb County, Missouri
##	1	1
##	Del Norte County, California	Delaware County, Iowa
##	1	1
##	Delaware County, Ohio	Delaware County, Oklahoma
##	1	1
##	Delaware County, Pennsylvania	Delta County, Michigan
##	1	1
##	Delta County, Texas	Dent County, Missouri
##	1	1
##	Denver County, Colorado	Des Moines County, Iowa
##	1	1
##	Deschutes County, Oregon	Desha County, Arkansas
##	1	1
##	Deuel County, South Dakota	Dewey County, Oklahoma
##	1	1
##	Dewey County, South Dakota	Dickenson County, Virginia
##	1	1
##	Dickey County, North Dakota	Dickinson County, Michigan
##	1	1
##	Dickson County, Tennessee	Dillon County, South Carolina
##	1	1
##	Dinwiddie County, Virginia	Dixie County, Florida
##	1	1
##	Dixon County, Nebraska	Doddridge County, West Virginia
##	1	1
##	Dodge County, Minnesota	Dodge County, Nebraska
##	1	1
##	Dodge County, Wisconsin	Dolores County, Colorado
##	1	1
##	Doña Ana County, New Mexico	Doniphan County, Kansas
##	1	1
##	Donley County, Texas	Dooly County, Georgia
##	1	1
##	Dorchester County, South Carolina	Dougherty County, Georgia
##	1	1
##	Douglas County, Colorado	Douglas County, Illinois

##	1	1
##	Douglas County, Kansas	Douglas County, Minnesota
##	1	1
##	Douglas County, Missouri	Douglas County, Nebraska
##	1	1
##	Douglas County, Nevada	Douglas County, Washington
##	1	1
##	Drew County, Arkansas	Dubois County, Indiana
##	1	1
##	Dubuque County, Iowa	Duchesne County, Utah
##	1	1
##	Dukes County, Massachusetts	Dundy County, Nebraska
##	1	1
##	Dunklin County, Missouri	Dunn County, North Dakota
##	1	1
##	Dunn County, Wisconsin	Duplin County, North Carolina
##	1	1
##	Dutchess County, New York	Duval County, Florida
##	1	1
##	Duval County, Texas	Dyer County, Tennessee
##	1	1
##	Eagle County, Colorado	Early County, Georgia
##	1	1
##	East Baton Rouge Parish, Louisiana	East Feliciana Parish, Louisiana
##	1	1
##	Eastland County, Texas	Echols County, Georgia
##	1	1
##	Ector County, Texas	Eddy County, North Dakota
##	1	1
##	Edwards County, Kansas	Effingham County, Georgia
##	1	1
##	El Dorado County, California	El Paso County, Texas
##	1	1
##	Elbert County, Georgia	Elk County, Pennsylvania
##	1	1
##	Elkhart County, Indiana	Elko County, Nevada
##	1	1
##	Elliott County, Kentucky	Elmore County, Alabama
##	1	1
##	Elmore County, Idaho	Emanuel County, Georgia
##	1	1
##	Emery County, Utah	Emmet County, Michigan
##	1	1
##	Emmons County, North Dakota	Erath County, Texas
##	1	1
##	Erie County, New York	Erie County, Pennsylvania
##	1	1
##	Escambia County, Florida	Esmeralda County, Nevada
##	1	1
##	Essex County, Massachusetts	Essex County, New York
##	1	1
##	Etowah County, Alabama	Eureka County, Nevada
##	1	1
##	Fairbanks North Star Borough, Alaska	Fairfax County, Virginia

##	1	1
##	Fairfield County, Connecticut	Fairfield County, Ohio
##	1	1
##	Fairfield County, South Carolina	Fallon County, Montana
##	1	1
##	Fannin County, Georgia	Fannin County, Texas
##	1	1
##	Faribault County, Minnesota	Faulk County, South Dakota
##	1	1
##	Faulkner County, Arkansas	Fauquier County, Virginia
##	1	1
##	Fayette County, Indiana	Fayette County, Kentucky
##	1	1
##	Fayette County, Ohio	Fayette County, Pennsylvania
##	1	1
##	Fayette County, Tennessee	Fentress County, Tennessee
##	1	1
##	Finney County, Kansas	Fisher County, Texas
##	1	1
##	Flagler County, Florida	Fleming County, Kentucky
##	1	1
##	Florence County, Wisconsin	Floyd County, Indiana
##	1	1
##	Floyd County, Kentucky	Floyd County, Virginia
##	1	1
##	Fluvanna County, Virginia	Ford County, Illinois
##	1	1
##	Ford County, Kansas	Forsyth County, Georgia
##	1	1
##	Foster County, North Dakota	Franklin city, Virginia
##	1	1
##	Franklin County, Florida	Franklin County, Georgia
##	1	1
##	Franklin County, Illinois	Franklin County, Iowa
##	1	1
##	Franklin County, Kentucky	Franklin County, Missouri
##	1	1
##	Franklin County, Nebraska	Franklin County, North Carolina
##	1	1
##	Franklin County, Vermont	Franklin County, Virginia
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##	Franklin County, Washington	Frederick County, Maryland
##	1	1
##	Frederick County, Virginia	Fredericksburg city, Virginia
##	1	1
##	Freeborn County, Minnesota	Freestone County, Texas
##	1	1
##	Fremont County, Idaho	Fremont County, Wyoming
##	1	1
##	Fresno County, California	Frontier County, Nebraska
##	1	1
##	Fulton County, Arkansas	Fulton County, New York
##	1	1
##	Fulton County, Pennsylvania	Furnas County, Nebraska

##	1	1
##	Gage County, Nebraska	Gallatin County, Illinois
##	1	1
##	Gallatin County, Montana	Gallia County, Ohio
##	1	1
##	Galveston County, Texas	Garden County, Nebraska
##	1	1
##	Garfield County, Colorado	Garfield County, Oklahoma
##	1	1
##	Garrard County, Kentucky	Garrett County, Maryland
##	1	1
##	Gasconade County, Missouri	Gem County, Idaho
##	1	1
##	Genesee County, Michigan	Geneva County, Alabama
##	1	1
##	George County, Mississippi	Gibson County, Indiana
##	1	1
##	Giles County, Tennessee	Gillespie County, Texas
##	1	1
##	Gilliam County, Oregon	Gilmer County, West Virginia
##	1	1
##	Gladwin County, Michigan	Glascock County, Georgia
##	1	1
##	Glenn County, California	Gloucester County, Virginia
##	1	1
##	Glynn County, Georgia	Gogebic County, Michigan
##	1	1
##	Goliad County, Texas	Gonzales County, Texas
##	1	1
##	Goodhue County, Minnesota	Gordon County, Georgia
##	1	1
##	Goshen County, Wyoming	Gosper County, Nebraska
##	1	1
##	Grady County, Georgia	Grady County, Oklahoma
##	1	1
##	Graham County, North Carolina	Grand County, Colorado
##	1	1
##	Grand County, Utah	Grand Isle County, Vermont
##	1	1
##	Grant County, Arkansas	Grant County, Kansas
##	1	1
##	Grant County, Kentucky	Grant County, Minnesota
##	1	1
##	Grant County, Oklahoma	Grant County, Oregon
##	1	1
##	Grant County, West Virginia	Grant Parish, Louisiana
##	1	1
##	Granville County, North Carolina	Graves County, Kentucky
##	1	1
##	Gray County, Kansas	Grays Harbor County, Washington
##	1	1
##	Grayson County, Texas	Grayson County, Virginia
##	1	1
##	Greeley County, Kansas	Greeley County, Nebraska

##	1	1
##	Green County, Kentucky	Green County, Wisconsin
##	1	1
##	Green Lake County, Wisconsin	Greene County, Illinois
##	1	1
##	Greene County, Indiana	Greene County, Iowa
##	1	1
##	Greene County, Missouri	Greene County, North Carolina
##	1	1
##	Greene County, Pennsylvania	Greene County, Tennessee
##	1	1
##	Greene County, Virginia	Greenlee County, Arizona
##	1	1
##	Greenup County, Kentucky	Greenville County, South Carolina
##	1	1
##	Greenwood County, South Carolina	Gregg County, Texas
##	1	1
##	Grenada County, Mississippi	Griggs County, North Dakota
##	1	1
##	Grundy County, Illinois	Grundy County, Iowa
##	1	1
##	Grundy County, Missouri	Guadalupe County, New Mexico
##	1	1
##	Gulf County, Florida	Guthrie County, Iowa
##	1	1
##	Haakon County, South Dakota	Habersham County, Georgia
##	1	1
##	Haines Borough, Alaska	Hale County, Alabama
##	1	1
##	Hale County, Texas	Halifax County, North Carolina
##	1	1
##	Halifax County, Virginia	Hall County, Georgia
##	1	1
##	Hall County, Nebraska	Hall County, Texas
##	1	1
##	Hamblen County, Tennessee	Hamilton County, Illinois
##	1	1
##	Hamilton County, Indiana	Hamilton County, Iowa
##	1	1
##	Hamilton County, Kansas	Hamilton County, Nebraska
##	1	1
##	Hamilton County, New York	Hamilton County, Ohio
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##	Hancock County, Mississippi	Hancock County, Tennessee
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##	Hancock County, West Virginia	Hansford County, Texas

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##	Highlands County, Florida	Hill County, Texas
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##	Hillsborough County, New Hampshire	Hillsdale County, Michigan
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##	Hockley County, Texas	Hoke County, North Carolina
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##	Holmes County, Mississippi	Hood County, Texas
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##	Hood River County, Oregon	Hopewell city, Virginia

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##	Howell County, Missouri	Huerfano County, Colorado
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##	Hughes County, Oklahoma	Hughes County, South Dakota
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##	Humphreys County, Tennessee	Hunt County, Texas
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##	Huron County, Michigan	Hutchinson County, South Dakota
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##	Hutchinson County, Texas	Hyde County, North Carolina
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##	Imperial County, California	Independence County, Arkansas
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##	Judith Basin County, Montana	Juneau County, Wisconsin
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##	King and Queen County, Virginia	King William County, Virginia
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##	Kosciusko County, Indiana	La Paz County, Arizona
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##	La Salle County, Texas	Lac qui Parle County, Minnesota
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##	Limestone County, Alabama	Lincoln County, Arkansas

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##	Lorain County, Ohio	Los Alamos County, New Mexico
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##	Los Angeles County, California	Loudoun County, Virginia
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##	Louisa County, Virginia	Lowndes County, Alabama
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##	Mason County, Washington	Mason County, West Virginia
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##	Massac County, Illinois	Matanuska-Susitna Borough, Alaska
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##	Maui County, Hawaii	Maury County, Tennessee

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##	Maverick County, Texas	Mayes County, Oklahoma
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##	McDonald County, Missouri	McDowell County, North Carolina
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##	Meade County, Kansas	Meade County, Kentucky
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##	New Haven County, Connecticut	New Kent County, Virginia
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##	New London County, Connecticut	New Madrid County, Missouri
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##	Newaygo County, Michigan	Newport County, Rhode Island
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##	Niobrara County, Wyoming	Noble County, Indiana
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##	Onondaga County, New York	Onslow County, North Carolina
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##	Ouray County, Colorado	Overton County, Tennessee
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##	Owen County, Indiana	Ozaukee County, Wisconsin
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##	Park County, Wyoming	Parke County, Indiana

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##	Platte County, Nebraska	Platte County, Wyoming

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##	Pointe Coupee Parish, Louisiana	Polk County, Florida
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##	Queens County, New York	Quitman County, Georgia
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##	1	1
##	Rains County, Texas	Randall County, Texas
##	1	1
##	Randolph County, Alabama	Randolph County, Arkansas

##	1	1
##	Randolph County, Georgia	Randolph County, North Carolina
##	1	1
##	Randolph County, West Virginia	Rankin County, Mississippi
##	1	1
##	Ransom County, North Dakota	Rapides Parish, Louisiana
##	1	1
##	Rappahannock County, Virginia	Ravalli County, Montana
##	1	1
##	Rawlins County, Kansas	Ray County, Missouri
##	1	1
##	Real County, Texas	Red Willow County, Nebraska
##	1	1
##	Redwood County, Minnesota	Reeves County, Texas
##	1	1
##	Reno County, Kansas	Rensselaer County, New York
##	1	1
##	Renville County, North Dakota	Republic County, Kansas
##	1	1
##	Rhea County, Tennessee	Rice County, Kansas
##	1	1
##	Richardson County, Nebraska	Richland County, Illinois
##	1	1
##	Richland County, Ohio	Richland County, South Carolina
##	1	1
##	Richland Parish, Louisiana	Richmond County, New York
##	1	1
##	Richmond County, North Carolina	Rio Arriba County, New Mexico
##	1	1
##	Ripley County, Indiana	Roane County, Tennessee
##	1	1
##	Roane County, West Virginia	Roanoke County, Virginia
##	1	1
##	Roberts County, South Dakota	Robeson County, North Carolina
##	1	1
##	Rock County, Minnesota	Rock County, Wisconsin
##	1	1
##	Rock Island County, Illinois	Rockbridge County, Virginia
##	1	1
##	Rockingham County, North Carolina	Rockingham County, Virginia
##	1	1
##	Rockland County, New York	Rogers County, Oklahoma
##	1	1
##	Rooks County, Kansas	Roosevelt County, Montana
##	1	1
##	Roscommon County, Michigan	Roseau County, Minnesota
##	1	1
##	Ross County, Ohio	Rowan County, Kentucky
##	1	1
##	Runnels County, Texas	Rush County, Kansas
##	1	1
##	Rusk County, Texas	Rusk County, Wisconsin
##	1	1
##	Russell County, Alabama	Russell County, Kentucky

##	1	1
##	Russell County, Virginia	Rutherford County, North Carolina
##	1	1
##	Rutherford County, Tennessee	Sabine County, Texas
##	1	1
##	Sabine Parish, Louisiana	Sacramento County, California
##	1	1
##	Sagadahoc County, Maine	Saginaw County, Michigan
##	1	1
##	Salem city, Virginia	Salem County, New Jersey
##	1	1
##	Saline County, Illinois	Saline County, Nebraska
##	1	1
##	Salt Lake County, Utah	Saluda County, South Carolina
##	1	1
##	Sampson County, North Carolina	San Augustine County, Texas
##	1	1
##	San Benito County, California	San Jacinto County, Texas
##	1	1
##	San Juan County, Utah	San Juan County, Washington
##	1	1
##	San Luis Obispo County, California	San Mateo County, California
##	1	1
##	San Miguel County, Colorado	San Miguel County, New Mexico
##	1	1
##	San Patricio County, Texas	San Saba County, Texas
##	1	1
##	Sanborn County, South Dakota	Sanders County, Montana
##	1	1
##	Sandusky County, Ohio	Sanpete County, Utah
##	1	1
##	Santa Barbara County, California	Santa Clara County, California
##	1	1
##	Santa Cruz County, Arizona	Santa Cruz County, California
##	1	1
##	Santa Fe County, New Mexico	Santa Rosa County, Florida
##	1	1
##	Sauk County, Wisconsin	Saunders County, Nebraska
##	1	1
##	Sawyer County, Wisconsin	Schenectady County, New York
##	1	1
##	Schleicher County, Texas	Schley County, Georgia
##	1	1
##	Schoharie County, New York	Schuyler County, New York
##	1	1
##	Schuylkill County, Pennsylvania	Scioto County, Ohio
##	1	1
##	Scotland County, Missouri	Scott County, Arkansas
##	1	1
##	Scott County, Illinois	Scott County, Iowa
##	1	1
##	Scott County, Kansas	Scott County, Virginia
##	1	1
##	Scurry County, Texas	Searcy County, Arkansas

##	1	1
##	Sebastian County, Arkansas	Sedgwick County, Colorado
##	1	1
##	Seminole County, Georgia	Seminole County, Oklahoma
##	1	1
##	Seneca County, New York	Seneca County, Ohio
##	1	1
##	Sequatchie County, Tennessee	Sequoyah County, Oklahoma
##	1	1
##	Seward County, Kansas	Sharkey County, Mississippi
##	1	1
##	Shawano County, Wisconsin	Shawnee County, Kansas
##	1	1
##	Shelby County, Alabama	Shelby County, Illinois
##	1	1
##	Shelby County, Kentucky	Shelby County, Missouri
##	1	1
##	Shelby County, Tennessee	Shenandoah County, Virginia
##	1	1
##	Sheridan County, Kansas	Sheridan County, Montana
##	1	1
##	Sheridan County, North Dakota	Sherman County, Nebraska
##	1	1
##	Sherman County, Oregon	Shiawassee County, Michigan
##	1	1
##	Sierra County, New Mexico	Simpson County, Mississippi
##	1	1
##	Siskiyou County, California	Sitka City and Borough, Alaska
##	1	1
##	Skagit County, Washington	Skamania County, Washington
##	1	1
##	Smith County, Kansas	Smith County, Mississippi
##	1	1
##	Smith County, Tennessee	Smith County, Texas
##	1	1
##	Smyth County, Virginia	Snohomish County, Washington
##	1	1
##	Snyder County, Pennsylvania	Solano County, California
##	1	1
##	Somerset County, Maine	Somerset County, New Jersey
##	1	1
##	Somerset County, Pennsylvania	Sonoma County, California
##	1	1
##	Southampton County, Virginia	Spalding County, Georgia
##	1	1
##	Spartanburg County, South Carolina	Spencer County, Indiana
##	1	1
##	Spencer County, Kentucky	Spink County, South Dakota
##	1	1
##	St. Charles Parish, Louisiana	St. Clair County, Alabama
##	1	1
##	St. Clair County, Michigan	St. Francois County, Missouri
##	1	1
##	St. Helena Parish, Louisiana	St. James Parish, Louisiana

##	1		1
##	St. John the Baptist Parish, Louisiana		St. Johns County, Florida
##	1		1
##	St. Joseph County, Indiana		St. Landry Parish, Louisiana
##	1		1
##	St. Lawrence County, New York		St. Lucie County, Florida
##	1		1
##	St. Martin Parish, Louisiana		St. Mary Parish, Louisiana
##	1		1
##	St. Mary's County, Maryland		St. Tammany Parish, Louisiana
##	1		1
##	Stafford County, Kansas		Stafford County, Virginia
##	1		1
##	Stanley County, South Dakota		Stanly County, North Carolina
##	1		1
##	Stanton County, Nebraska		Starke County, Indiana
##	1		1
##	Starr County, Texas		Staunton city, Virginia
##	1		1
##	Ste. Genevieve County, Missouri		Stearns County, Minnesota
##	1		1
##	Steele County, Minnesota		Steele County, North Dakota
##	1		1
##	Stephens County, Georgia		Stephens County, Oklahoma
##	1		1
##	Stephens County, Texas		Steuben County, Indiana
##	1		1
##	Stevens County, Kansas		Stevens County, Washington
##	1		1
##	Stewart County, Georgia		Stillwater County, Montana
##	1		1
##	Stoddard County, Missouri		Stone County, Missouri
##	1		1
##	Stonewall County, Texas		Storey County, Nevada
##	1		1
##	Story County, Iowa		Strafford County, New Hampshire
##	1		1
##	Stutsman County, North Dakota		Suffolk city, Virginia
##	1		1
##	Suffolk County, Massachusetts		Sullivan County, Missouri
##	1		1
##	Sullivan County, New York		Sullivan County, Pennsylvania
##	1		1
##	Sullivan County, Tennessee		Sully County, South Dakota
##	1		1
##	Summers County, West Virginia		Summit County, Utah
##	1		1
##	Sumner County, Kansas		Sumner County, Tennessee
##	1		1
##	Sumter County, Georgia		Sunflower County, Mississippi
##	1		1
##	Surry County, Virginia		Susquehanna County, Pennsylvania
##	1		1
##	Suwannee County, Florida		Sweetwater County, Wyoming

##	1	1
##	Swift County, Minnesota	Swisher County, Texas
##	1	1
##	Switzerland County, Indiana	Talbot County, Maryland
##	1	1
##	Taliaferro County, Georgia	Tallahatchie County, Mississippi
##	1	1
##	Tama County, Iowa	Taos County, New Mexico
##	1	1
##	Tate County, Mississippi	Taylor County, Florida
##	1	1
##	Taylor County, Iowa	Taylor County, Wisconsin
##	1	1
##	Terrebonne Parish, Louisiana	Terry County, Texas
##	1	1
##	Teton County, Idaho	Texas County, Oklahoma
##	1	1
##	Thayer County, Nebraska	Thomas County, Georgia
##	1	1
##	Throckmorton County, Texas	Thurston County, Nebraska
##	1	1
##	Tift County, Georgia	Tioga County, Pennsylvania
##	1	1
##	Tippah County, Mississippi	Tippecanoe County, Indiana
##	1	1
##	Tipton County, Indiana	Tishomingo County, Mississippi
##	1	1
##	Todd County, Minnesota	Tolland County, Connecticut
##	1	1
##	Tompkins County, New York	Tooele County, Utah
##	1	1
##	Toombs County, Georgia	Torrance County, New Mexico
##	1	1
##	Towns County, Georgia	Traill County, North Dakota
##	1	1
##	Traverse County, Minnesota	Trego County, Kansas
##	1	1
##	Trigg County, Kentucky	Tripp County, South Dakota
##	1	1
##	Trousdale County, Tennessee	Trumbull County, Ohio
##	1	1
##	Tucker County, West Virginia	Tulsa County, Oklahoma
##	1	1
##	Tuolumne County, California	Turner County, Georgia
##	1	1
##	Turner County, South Dakota	Tuscaloosa County, Alabama
##	1	1
##	Tuscarawas County, Ohio	Twiggs County, Georgia
##	1	1
##	Twin Falls County, Idaho	Tyler County, Texas
##	1	1
##	Tyrrell County, North Carolina	Umatilla County, Oregon
##	1	1
##	Unicoi County, Tennessee	Union County, Florida

##	1	1
##	Union County, Georgia	Union County, Iowa
##	1	1
##	Union County, Kentucky	Union County, New Jersey
##	1	1
##	Union County, New Mexico	Union County, North Carolina
##	1	1
##	Union County, Ohio	Union County, Oregon
##	1	1
##	Union County, Pennsylvania	Union County, South Carolina
##	1	1
##	Union County, South Dakota	Upson County, Georgia
##	1	1
##	Upton County, Texas	Utah County, Utah
##	1	1
##	Uvalde County, Texas	Val Verde County, Texas
##	1	1
##	Valencia County, New Mexico	Valley County, Idaho
##	1	1
##	Van Buren County, Michigan	Van Buren County, Tennessee
##	1	1
##	Van Wert County, Ohio	Van Zandt County, Texas
##	1	1
##	Vermilion County, Illinois	Vermilion Parish, Louisiana
##	1	1
##	Vermillion County, Indiana	Vernon County, Missouri
##	1	1
##	Vernon County, Wisconsin	Vernon Parish, Louisiana
##	1	1
##	Vigo County, Indiana	Vilas County, Wisconsin
##	1	1
##	Virginia Beach city, Virginia	Wabash County, Illinois
##	1	1
##	Wabaunsee County, Kansas	Wadena County, Minnesota
##	1	1
##	Wagoner County, Oklahoma	Wake County, North Carolina
##	1	1
##	Wakulla County, Florida	Walla Walla County, Washington
##	1	1
##	Walsh County, North Dakota	Walthall County, Mississippi
##	1	1
##	Walton County, Georgia	Walworth County, South Dakota
##	1	1
##	Wapello County, Iowa	Ward County, North Dakota
##	1	1
##	Ware County, Georgia	Warren County, Illinois
##	1	1
##	Warren County, Indiana	Warren County, Iowa
##	1	1
##	Warren County, Kentucky	Warren County, Mississippi
##	1	1
##	Warren County, New Jersey	Warren County, North Carolina
##	1	1
##	Warren County, Ohio	Warren County, Pennsylvania

##	1	1
##	Warrick County, Indiana	Wasatch County, Utah
##	1	1
##	Washakie County, Wyoming	Washburn County, Wisconsin
##	1	1
##	Washington County, Alabama	Washington County, Arkansas
##	1	1
##	Washington County, Colorado	Washington County, Idaho
##	1	1
##	Washington County, Illinois	Washington County, Indiana
##	1	1
##	Washington County, Kansas	Washington County, Kentucky
##	1	1
##	Washington County, Maryland	Washington County, Minnesota
##	1	1
##	Washington County, Missouri	Washington County, Nebraska
##	1	1
##	Washington County, North Carolina	Washington County, Ohio
##	1	1
##	Washington County, Texas	Washington County, Virginia
##	1	1
##	Washington County, Wisconsin	Washington Parish, Louisiana
##	1	1
##	Washita County, Oklahoma	Waushara County, Wisconsin
##	1	1
##	Wayne County, Georgia	Wayne County, Illinois
##	1	1
##	Wayne County, Iowa	Wayne County, Michigan
##	1	1
##	Wayne County, Mississippi	Wayne County, Missouri
##	1	1
##	Wayne County, New York	Wayne County, Ohio
##	1	1
##	Wayne County, Pennsylvania	Wayne County, Tennessee
##	1	1
##	Wayne County, West Virginia	Waynesboro city, Virginia
##	1	1
##	Weakley County, Tennessee	Webb County, Texas
##	1	1
##	Webster County, Missouri	Webster County, Nebraska
##	1	1
##	Webster County, West Virginia	Webster Parish, Louisiana
##	1	1
##	Weld County, Colorado	Wells County, Indiana
##	1	1
##	Wells County, North Dakota	West Carroll Parish, Louisiana
##	1	1
##	Westchester County, New York	Wetzel County, West Virginia
##	1	1
##	Wharton County, Texas	Whatcom County, Washington
##	1	1
##	Wheatland County, Montana	Wheeler County, Oregon
##	1	1
##	White County, Illinois	White Pine County, Nevada

##	1	1
##	Whiteside County, Illinois	Whitfield County, Georgia
##	1	1
##	Whitley County, Indiana	Whitley County, Kentucky
##	1	1
##	Whitman County, Washington	Wichita County, Kansas
##	1	1
##	Wilbarger County, Texas	Wilkes County, Georgia
##	1	1
##	Wilkes County, North Carolina	Wilkin County, Minnesota
##	1	1
##	Wilkinson County, Georgia	Wilkinson County, Mississippi
##	1	1
##	Will County, Illinois	Willacy County, Texas
##	1	1
##	Williams County, North Dakota	Williams County, Ohio
##	1	1
##	Williamsburg city, Virginia	Williamsburg County, South Carolina
##	1	1
##	Williamson County, Illinois	Williamson County, Tennessee
##	1	1
##	Wilson County, Kansas	Wilson County, Texas
##	1	1
##	Windham County, Vermont	Winkler County, Texas
##	1	1
##	Winn Parish, Louisiana	Winnebago County, Illinois
##	1	1
##	Wise County, Texas	Wise County, Virginia
##	1	1
##	Wolfe County, Kentucky	Wood County, Ohio
##	1	1
##	Wood County, West Virginia	Wood County, Wisconsin
##	1	1
##	Woodford County, Kentucky	Woodruff County, Arkansas
##	1	1
##	Woods County, Oklahoma	Woodward County, Oklahoma
##	1	1
##	Worcester County, Massachusetts	Worth County, Missouri
##	1	1
##	Wright County, Minnesota	Wyandot County, Ohio
##	1	1
##	Wyandotte County, Kansas	Wyoming County, Pennsylvania
##	1	1
##	Yamhill County, Oregon	Yancey County, North Carolina
##	1	1
##	Yankton County, South Dakota	Yates County, New York
##	1	1
##	Yell County, Arkansas	Yellowstone County, Montana
##	1	1
##	Yoakum County, Texas	Yolo County, California
##	1	1
##	York County, Nebraska	York County, Pennsylvania
##	1	1
##	York County, South Carolina	York County, Virginia

```
##                                1                                1
##          Yuba County, California      Yukon-Koyukuk Census Area, Alaska
##                                1                                1
##          Yuma County, Arizona          Yuma County, Colorado
##                                1                                1
##          Zavala County, Texas
##                                1
```

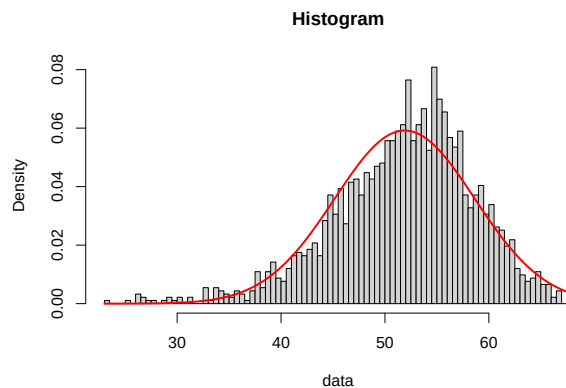
Variable *percentmarried*

Numerical variable that represent the percent of county residents who are married.

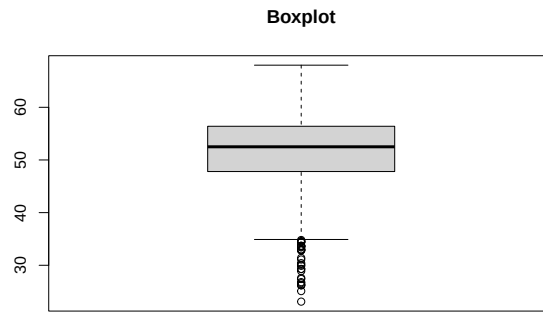
It has a non-symmetric distribution, it is skewed with a long left tail observed in the histogram. It doesn't have missing values, it has 34 outliers (non of them severe) and they are concentrated on the lower end of the distribution. With the shapiro-test we can confirm the rejection of the null hypothesis, therefore it doesn't have a normal distribution.

```
variable <- train$percentmarried
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      23.1   47.8    52.5    51.9   56.4    68.0
## [1] "Mean"
## [1] 51.90087
## [1] "Standard deviation"
## [1] 6.740129
```



```
## [1] "Skewness"
## [1] -0.6474867
```



```
## [1] "Num Outliers:"
## [1] 34
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.97753, p-value = 2.346e-16
outliers_df[res.var,"percentmarried"] <- 1
```

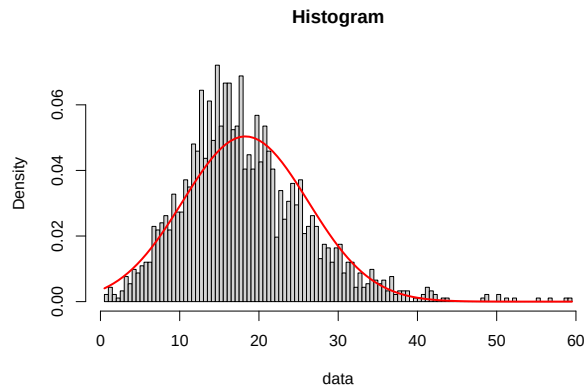
Variable *pctnohs18_24*

Numerical variable that show the percentage of county residents aged 18-24 whose highest level of education is less than a high school diploma.

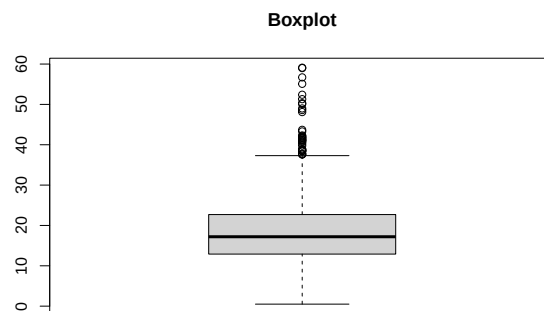
The distribution of this variable is skewed with a long right tail, we can confirm that it doesn't have a normal distribution thanks to the histogram and the curve. Moreover the p-value obtained in the shapiro-test confirm the non normal distribution. It doesn't have missing values, and it has 35 outliers, 5 of them severe.

```
variable <- train$pctnohs18_24
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   0.50  12.90   17.20   18.29  22.70   59.10
## [1] "Mean"
## [1] 18.28777
## [1] "Standard deviation"
## [1] 7.92004
```



```
## [1] "Skewness"
## [1] 0.8925103
```



```
## [1] "Num Outliers:"
## [1] 35
## [1] "Num. severe outliers:"
## [1] 5
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.96205, p-value < 2.2e-16
outliers_df[res.var,"pctnohs18_24"] <- 1
```

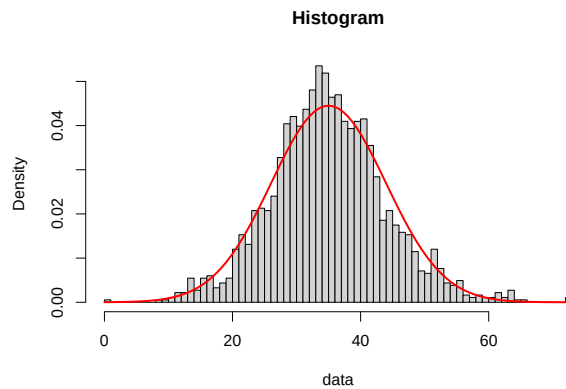
Variable *ppcths18_24*

Numeric variable, it represent the percentage of county residents ages 18-24 highest education attained is a high school diploma.

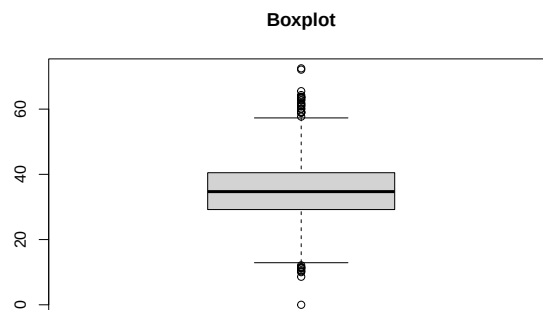
We can observe with the median(34,7%) and the mean(35%) that the variable has a symmetric distribution, but after the shapiro_test we observe that it doesn't have a normal distribution. It doesn't have missing values and it has 33 outliers, non of them severe.

```
variable <- train$ppcths18_24
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.0   29.2   34.7   35.0   40.5   72.5
## [1] "Mean"
## [1] 34.99978
## [1] "Standard deviation"
## [1] 8.964462
```



```
## [1] "Skewness"
## [1] 0.2205398
```



```
## [1] "Num Outliers:"
## [1] 33
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
##  Shapiro-Wilk normality test
##
## data:  data
## W = 0.99323, p-value = 1.922e-07
outliers_df[res.var,"pcths18_24"] <- 1
```

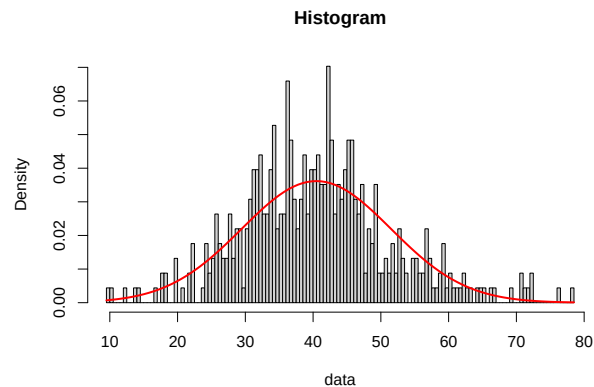
Variable *pctsomecol18_24* NOT WORTH TO USE IT (75% MISSING VALUES)

Numerical variable. Percent of county residents ages 18-24 highest education attained some college.

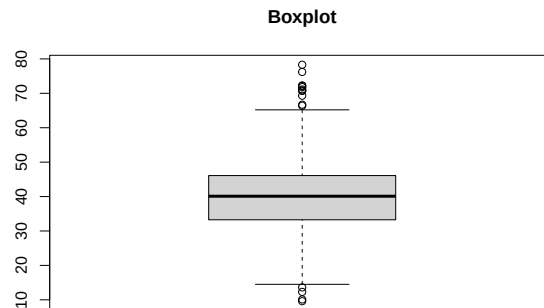
It has 1375 missing values. (75%) therefore we would remove this variable. We could obtain the values knowing that $pctnohs18_24 + pcths18_24 + pctsomecol18_24 + pctbachdeg18_24 = 100$

```
variable <- train$pctsomecol18_24
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##   9.60  33.25   40.10   40.48  46.10   78.30   1376
## [1] "Mean"
## [1] 40.47846
## [1] "Standard deviation"
## [1] 11.0307
```



```
## [1] "Skewness"
## [1] 0.4010929
```



```
## [1] "Num Outliers:"
## [1] 1391
## [1] "Num. severe outliers:"
## [1] 1376
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data:  data
## W = 0.98569, p-value = 0.0001868
```

```
outliers_df[res.var,"pctsomecol18_24"] <- 1
outliers <- which(outliers_df$pctsomecol18_24 == 1); length(outliers)
```

```
## [1] 15
```

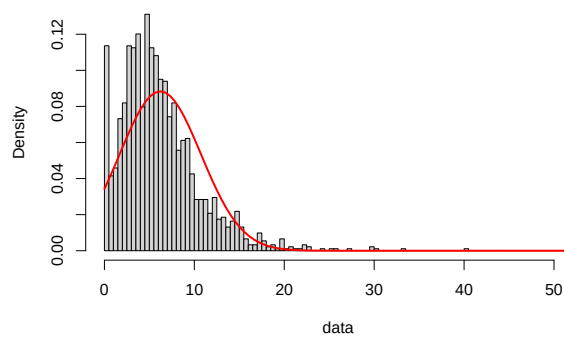
Variable *pctbachdeg18_24*

Numerical variable. Percent of county residents ages 18-24 highest education attained is a bachelor's degree. There are no missing values, it has 56 mild outliers 10 of them severes. After the shapiro-test we can reject the null hypothesis, thus it doesn't have a normal distribution.

```
variable <- train$pctbachdeg18_24
res.var <- summary_and_normality_check(variable)
```

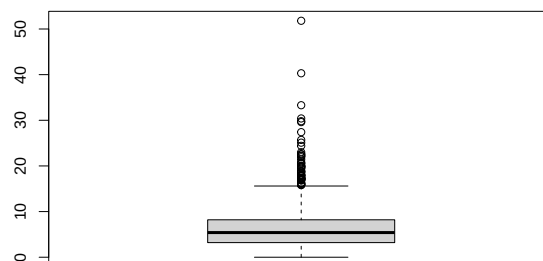
```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    0.000  3.200   5.400   6.216  8.200  51.800
## [1] "Mean"
## [1] 6.216221
## [1] "Standard deviation"
## [1] 4.517556
```

Histogram



```
## [1] "Skewness"
## [1] 1.980584
```

Boxplot



```
## [1] "Num Outliers:"
## [1] 56
```

```
## [1] "Num. severe outliers:"
## [1] 10
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.87959, p-value < 2.2e-16
outliers_df[res.var,"pctbachdeg18_24"] <- 1
```

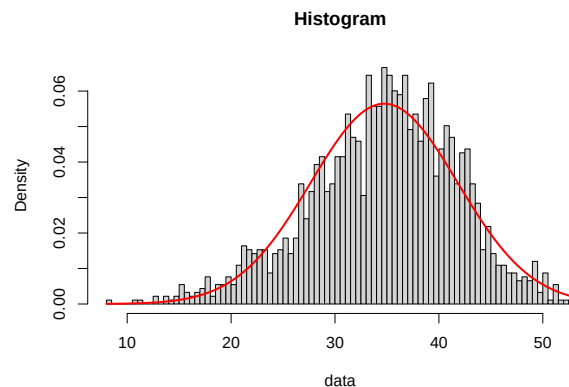
Variable *pcths25_over*

Numerical variable that express the percentage of county residents ages 25 and over highest education attained a high school diploma.

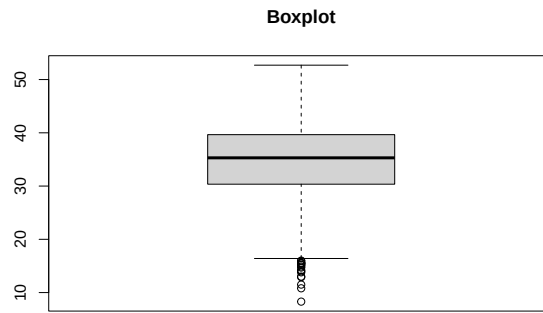
It doesn't have missing values but it have 18 outliers, non of the severe. It doesn't have a normal distribution.

```
variable <- train$pcths25_over
res.var <- summary_and_normality_check(variable)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   8.30  30.35   35.30   34.73   39.65   52.70
## [1] "Mean"
## [1] 34.73255
## [1] "Standard deviation"
## [1] 7.06599
```



```
## [1] "Skewness"
## [1] -0.3660496
```



```
## [1] "Num Outliers:"
## [1] 18
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99107, p-value = 3.741e-09

outliers_df[res.var,"pcths25_over"] <- 1
outliers <- which(outliers_df$pcths25_over == 1); length(outliers)

## [1] 18
```

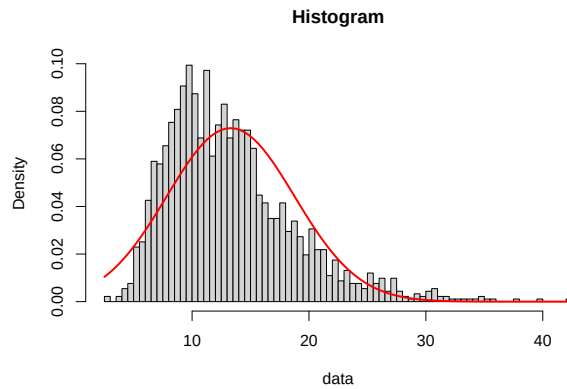
Variable *pctbachdeg25_over*

Numerical variable that expresses the the percentage of country resident ages 25 and over highest education attained: Bachelor's degree.

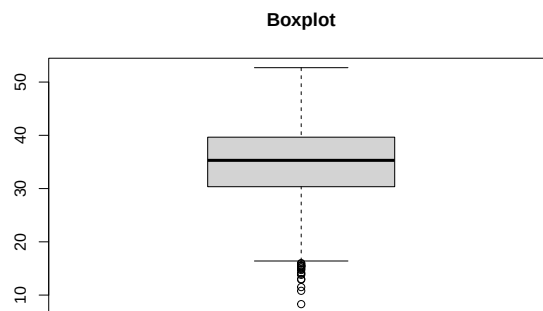
Firstly we can see that there is a huge left skewness and there is a relatively large number of outliers (62), from which just 3 are severe. From the plot we can already tell that no normal distribution is followed and a shapiro test over the variable confirms it.

```
res.var <- summary_and_normality_check(train$pctbachdeg25_over)

## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   2.5    9.3    12.3    13.3    16.0    42.2
## [1] "Mean"
## [1] 13.29754
## [1] "Standard deviation"
## [1] 5.470341
```



```
## [1] "Skewness"
## [1] 1.159947
```



```
## [1] "Num Outliers:"
## [1] 59
## [1] "Num. severe outliers:"
## [1] 3
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.92998, p-value < 2.2e-16
outliers_df[res.var,"pctbachdeg25_over"] <- 1
outliers <- which(outliers_df$pctbachdeg25_over == 1); length(outliers)

## [1] 59
```

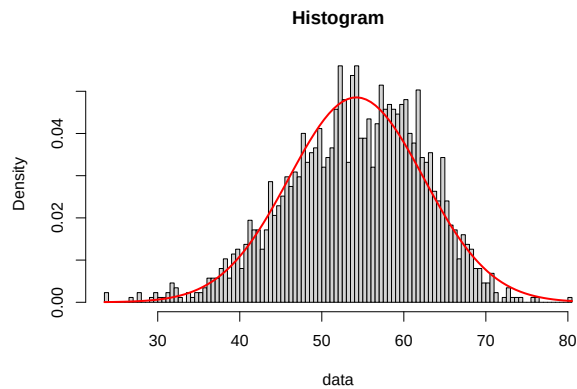
Variable *pctemployed16_over*

Numerical variable that represents the percentage of county residents ages 16 and over, employed.

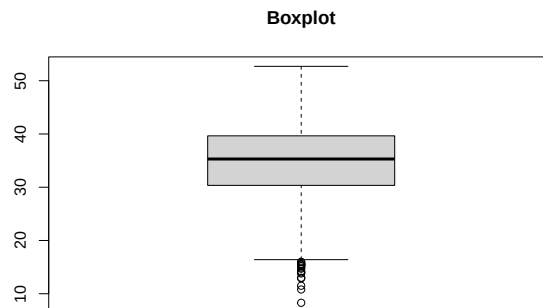
This variable contains over 82 NA values and has a slight right skewness and 11 mild outliers. Both inferential and visualization methods clearly point that the real distribution is extremely unlikely to be normal.

```
res.var <- summary_and_normality_check(train$pctemployed16_over)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
##      23.90  48.60   54.50   54.21  60.30   80.10      82
## [1] "Mean"
## [1] 54.21172
## [1] "Standard deviation"
## [1] 8.216153
```



```
## [1] "Skewness"
## [1] -0.3362687
```



```
## [1] "Num Outliers:"
## [1] 93
## [1] "Num. severe outliers:"
## [1] 82
## [1] "Shapiro test: "
##
##  Shapiro-Wilk normality test
##
## data:  data
## W = 0.99196, p-value = 3.371e-08
res.var
```

```
## [1] 434 720 723 753 1091 1138 1468 1547 1615 1633 1736
outliers_df[res.var,"pctemployed16_over"] <- 1
outliers <- which(outliers_df$pctemployed16_over == 1); length(outliers)
```

```
## [1] 11
```

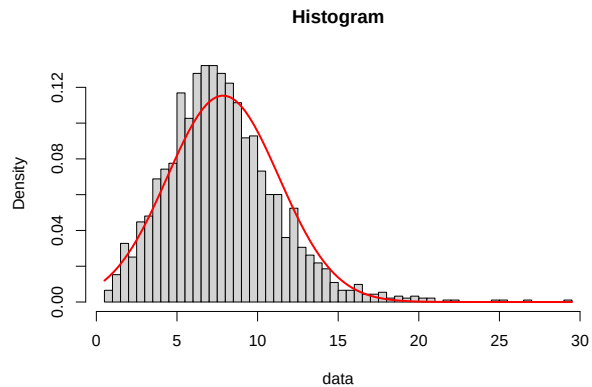
Variable *pctunemployed16_over*

Numerical variable that represents the percentage of county residents ages 16 and over, unemployed.

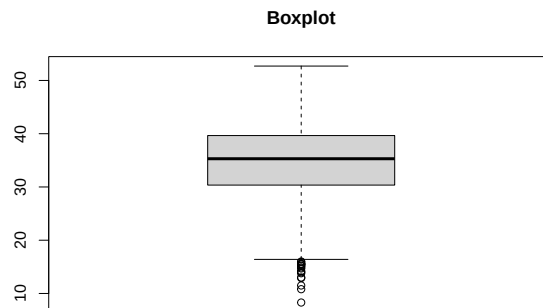
This variable is heavily left skewed and contains over 46 outliers, of which 4 are severe. Despite the distribution looking almost normal there is a big number of outliers and a long heavy right tail. Applying a shapiro test we reject the hypothesis that the distribution is normal.

```
res.var <- summary_and_normality_check(train$pctunemployed16_over)
```

```
## [1] "Summary: "  
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##  0.700  5.500   7.500   7.861  9.750  29.400  
## [1] "Mean"  
## [1] 7.86095  
## [1] "Standard deviation"  
## [1] 3.457111
```



```
## [1] "Skewness"  
## [1] 0.9189968
```



```
## [1] "Num Outliers:"  
## [1] 42  
## [1] "Num. severe outliers:"  
## [1] 4  
## [1] "Shapiro test: "
```

```
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.9612, p-value < 2.2e-16

outliers_df[res.var,"pctunemployed16_over"] <- 1
outliers <- which(outliers_df$pctunemployed16_over == 1); length(outliers)

## [1] 42
```

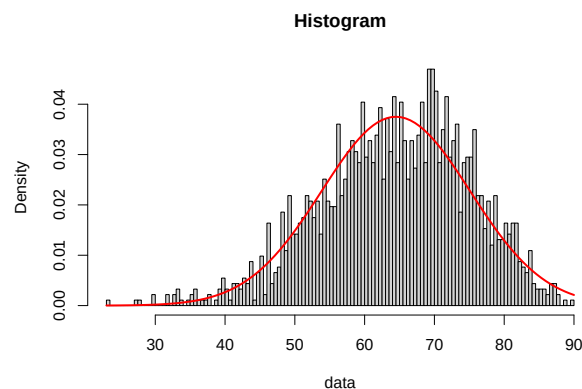
Variable *pctprivatecoverage*

Numerical variable that represents the percentage of county residents with private health coverage.

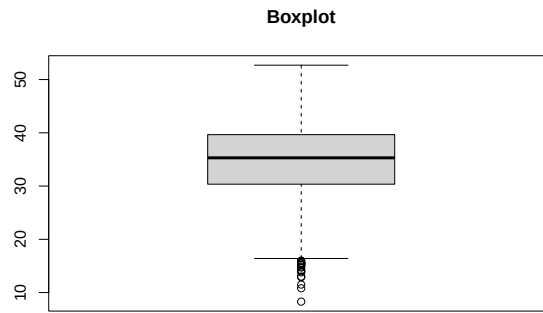
Despite having no NA and just 17 mild outliers, this variable has a huge standard deviation and a right skewed mean. Applying a shapiro test we reject the hypothesis that the distribution is normal.

```
res.var <- summary_and_normality_check(train$pctprivatecoverage)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  23.40  57.50   65.20   64.47   72.10   89.60
## [1] "Mean"
## [1] 64.4746
## [1] "Standard deviation"
## [1] 10.63614
```



```
## [1] "Skewness"
## [1] -0.3796678
```



```
## [1] "Num Outliers:"
## [1] 17
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98964, p-value = 3.725e-10

outliers_df[res.var,"pctprivatecoverage"] <- 1
outliers <- which(outliers_df$pctprivatecoverage == 1); length(outliers)

## [1] 17
```

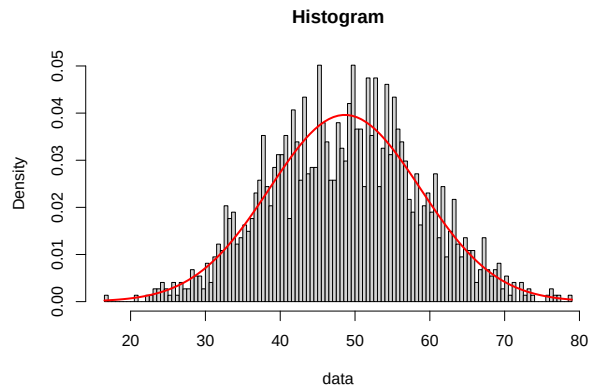
Variable *pctprivatecoveragealone*

Numerical variable that represents the percentage of county residents with private health coverage alone (no public assistance at all).

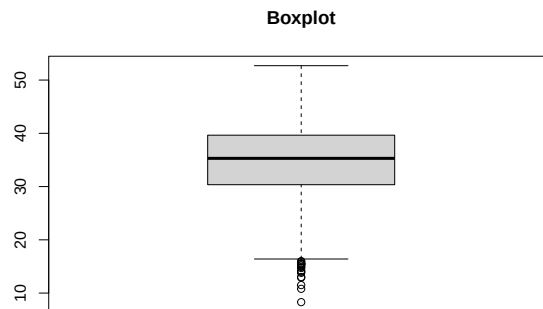
The number of NA is humongous (356), but it has a small amount of mild outliers (4) and almost no skewness at all. Both using inferential and visualizations methods we cannot easily reject the hypothesis that this variable is not normal. So we can loosely say that is **normally** distributed.

```
res.var <- summary_and_normality_check(train$pctprivatecoveragealone)

## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   NA's
##    16.80  41.50   49.00   48.65  55.50   78.90   356
## [1] "Mean"
## [1] 48.64807
## [1] "Standard deviation"
## [1] 10.06977
```



```
## [1] "Skewness"
## [1] 0.01553065
```



```
## [1] "Num Outliers:"
## [1] 360
## [1] "Num. severe outliers:"
## [1] 356
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99831, p-value = 0.1453
res.var
```

```
## [1] 106 128 636 1472
```

```
outliers_df[res.var,"pctprivatecoveragealone"] <- 1
outliers <- which(outliers_df$pctprivatecoveragealone == 1); length(outliers)
```

```
## [1] 0
```

Variable *pctempprivcoverage*

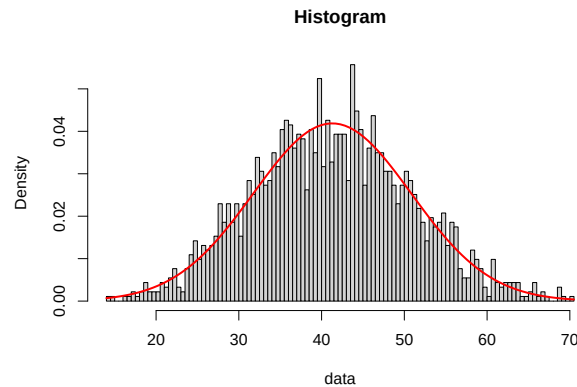
Numerical variable that represents the percentage of county residents with employee-provided private health coverage.

Despite the low number of outliers (7 mild), the absence of NA, the small left skewness and the median being

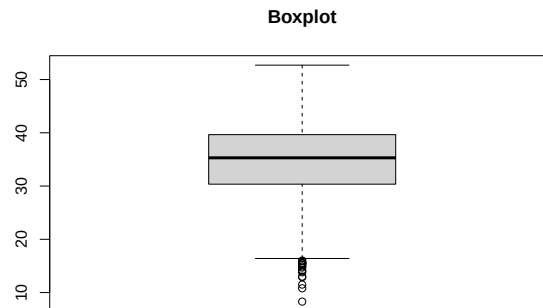
almost equal to the mean, we should reject the null hypothesis that the distribution is normal with a 5% significance. However, with a smaller significance level than 2.5% we couldn't reject that this distribution is not normal. **Almost normal**

```
res.var <- summary_and_normality_check(train$pctempprivcoverage)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##    14.30  34.60   41.10   41.29  47.70   70.20
## [1] "Mean"
## [1] 41.29388
## [1] "Standard deviation"
## [1] 9.528869
```



```
## [1] "Skewness"
## [1] 0.1082829
```



```
## [1] "Num Outliers:"
## [1] 7
## [1] "Num. severe outliers:"
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.99807, p-value = 0.02861
```

```
outliers_df[res.var,"pctempprivcoverage"] <- 1
```

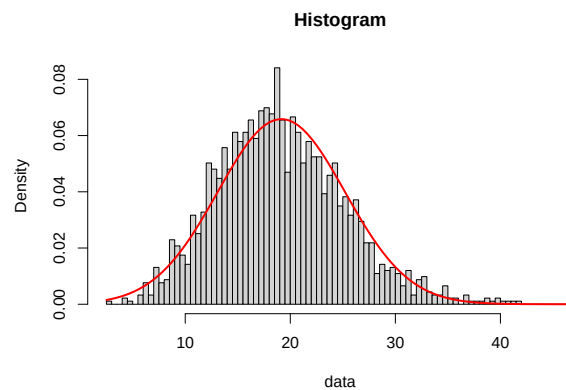
Variable *pctpubliccoveragealone*

Numerical variable that represents the percent of county residents with government-provided health coverage alone.

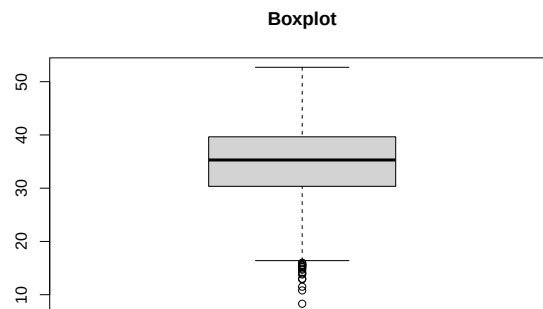
This variable contains 21 mild outliers, a great left skewness and a long right tail. Performing an inferential test like shapiro we get a tiny p-value which makes us reject the hypothesis that the distribution is normal under any significance level.

```
res.var <- summary_and_normality_check(train$pctpubliccoveragealone)
```

```
## [1] "Summary: "  
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##   2.60  14.90   18.70   19.15  23.00   46.60  
## [1] "Mean"  
## [1] 19.14992  
## [1] "Standard deviation"  
## [1] 6.056619
```



```
## [1] "Skewness"  
## [1] 0.4695236
```



```
## [1] "Num Outliers:"  
## [1] 21  
## [1] "Num. severe outliers:"
```

```
## [1] 0
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.98784, p-value = 2.648e-11
outliers_df[res.var,"pctpubliccoveragealone"] <- 1
```

Variable *pctwhite*

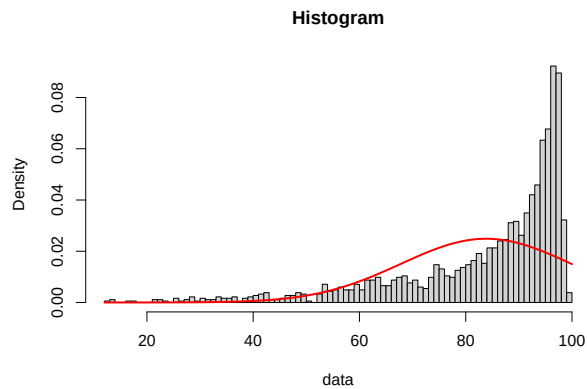
Numerical variable that represents the percent of county residents who identify as White.

Both from the plot and from inferential methods we can easily discard that this variable is anywhere near normal, being a hugely right skewed variable. It contains 104 outliers from which 7 are severe.

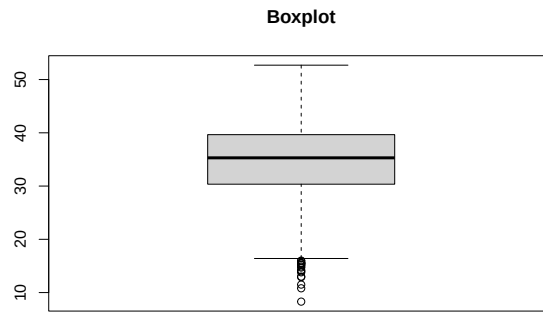
This points out that the vast majority of counties have a high percentage of white population.

```
res.var <- summary_and_normality_check(train$pctwhite)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  12.27   77.31   89.90   83.85   95.57   99.69
## [1] "Mean"
## [1] 83.84699
## [1] "Standard deviation"
## [1] 16.03708
```



```
## [1] "Skewness"
## [1] -1.625604
```



```
## [1] "Num Outliers:"
## [1] 97
## [1] "Num. severe outliers:"
## [1] 7
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.80758, p-value < 2.2e-16
outliers_df[res.var,"pctwhite"] <- 1
```

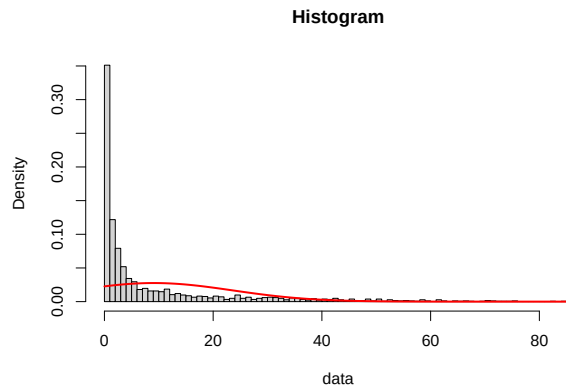
Variable *pctblack*

Numerical variable that represents the percent of county residents who identify as Black.

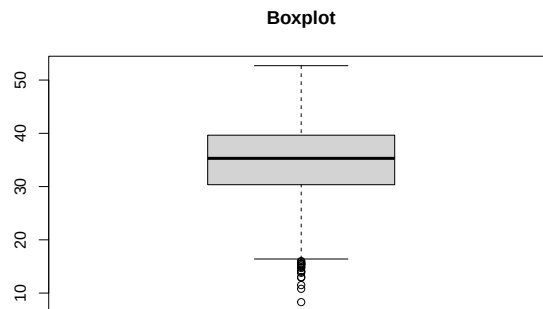
There is a huge number of outliers (323), from which 99 are severe and there is also a huge left skewness, that points out that the vast majority of counties (except the outliers) have a rather low percentage of black folks. The distribution almost looks like a power law distribution (Zipf) therefore we can safely reject both with visualization and inferential methods that the distribution of this variable is not normal.

```
res.var <- summary_and_normality_check(train$pctblack)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      0.000  0.648   2.323   9.082  10.867  85.948
## [1] "Mean"
## [1] 9.081563
## [1] "Standard deviation"
## [1] 14.45309
```



```
## [1] "Skewness"
## [1] 2.272138
```



```
## [1] "Num Outliers:"
## [1] 224
## [1] "Num. severe outliers:"
## [1] 99
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.65926, p-value < 2.2e-16
outliers_df[res.var,"pctblack"] <- 1
```

Variable *pctasian*

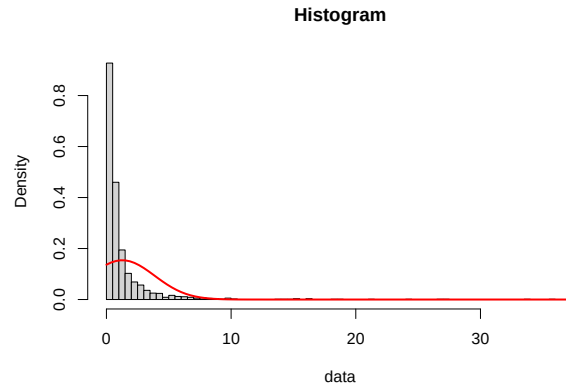
Numerical variable that represents the percent of county residents who identify as Asian.

Pretty similarly to the last distribution, we have a heavily left skewed zipfian like distribution, with 305 outliers (from which 107 are severe), pointing out that the vast majority of counties have less than 3% of asian percentage.

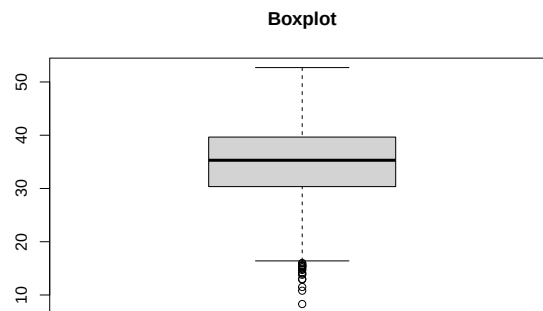
```
res.var <- summary_and_normality_check(train$pctasian)
```

```
## [1] "Summary: "
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

```
## 0.0000 0.2582 0.5495 1.2743 1.2515 37.1569
## [1] "Mean"
## [1] 1.274301
## [1] "Standard deviation"
## [1] 2.588845
```



```
## [1] "Skewness"
## [1] 7.103922
```



```
## [1] "Num Outliers:"
## [1] 198
## [1] "Num. severe outliers:"
## [1] 107
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.41908, p-value < 2.2e-16
```

```
outliers_df[res.var,"pctasian"] <- 1
```

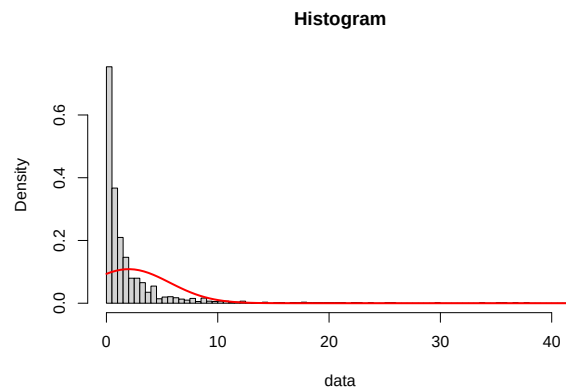
Variable *pctotherrace*

Numerical variable that represents the percent of county residents who identify as any other race (Pacific islanders, arábians, saiayajins, namekians...).

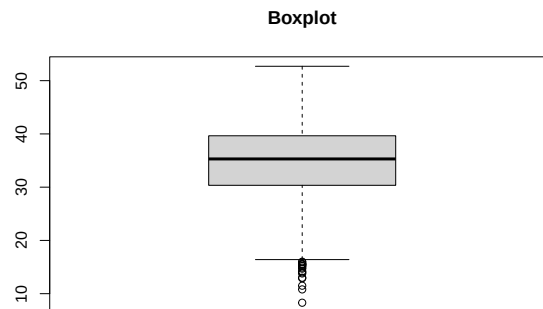
Again this variable, just like the last 2 presents a huge left skewness and zipfian like distribution with 283 outliers, from which 102 are severe. Again this sampled distribution is not normal.

```
res.var <- summary_and_normality_check(train$pctotherrace)
```

```
## [1] "Summary: "
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.2867  0.7826  2.0031  2.1066 41.9303
## [1] "Mean"
## [1] 2.003134
## [1] "Standard deviation"
## [1] 3.672809
```



```
## [1] "Skewness"
## [1] 4.987083
```



```
## [1] "Num Outliers:"
## [1] 181
## [1] "Num. severe outliers:"
## [1] 102
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.50981, p-value < 2.2e-16
```

```
outliers_df[res.var,"pctotherrace"] <- 1
```

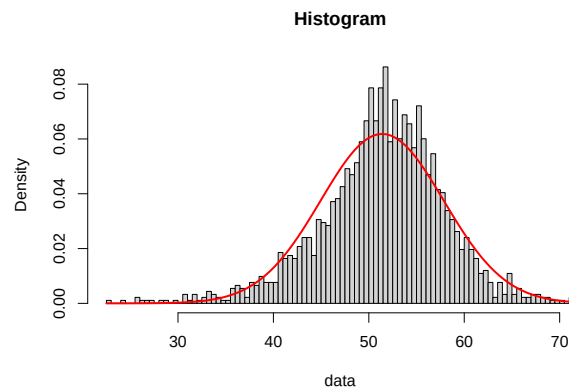
Variable *pctmarriedhouseholds*

Numerical variable that represents the percent of married households.

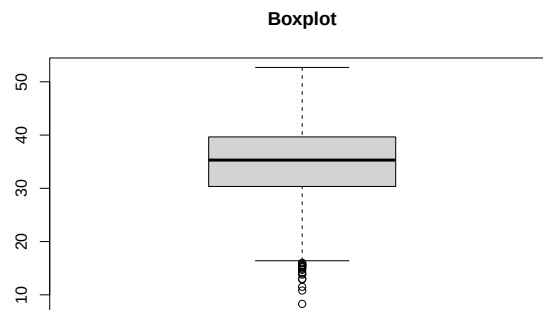
This variable contains 59 outliers from which 2 are severe and a noticeable left skewness. Visualization isn't that clear, but using inferential methods like shapiro test we reject the null hypothesis that this variable is normal.

```
res.var <- summary_and_normality_check(train$pctmarriedhouseholds)
```

```
## [1] "Summary: "  
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
##  22.99  47.85   51.73   51.40  55.48   71.40  
## [1] "Mean"  
## [1] 51.40232  
## [1] "Standard deviation"  
## [1] 6.450696
```



```
## [1] "Skewness"  
## [1] -0.5031018
```



```
## [1] "Num Outliers:"  
## [1] 57  
## [1] "Num. severe outliers:"  
## [1] 2
```

```
## [1] "Shapiro test: "  
##  
## Shapiro-Wilk normality test  
##  
## data: data  
## W = 0.9816, p-value = 1.341e-14
```

```
outliers_df[res.var,"pctmarriedhouseholds"] <- 1
```

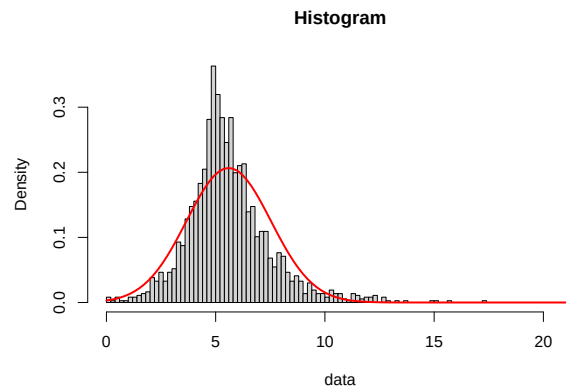
Variable *birthrate*

Numerical variable that represents the number of births relative to number of women in that specific county.

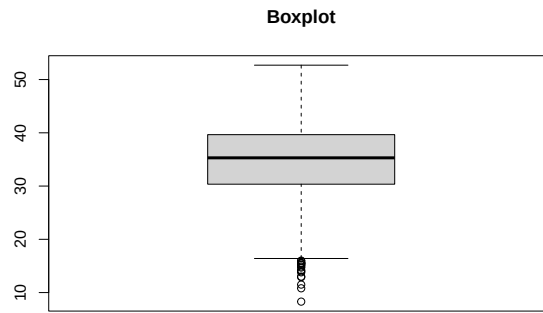
Finally, this last variable contains a fair amount of outliers (121) from which 17 are severe. It also has a fair amount of left skewness. Both from visualization and shapiro testing we reject that this sample variable is normal.

```
res.var <- summary_and_normality_check(train$birthrate)
```

```
## [1] "Summary: "  
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.  
## 0.000  4.528   5.355   5.597   6.414   21.326  
## [1] "Mean"  
## [1] 5.597439  
## [1] "Standard deviation"  
## [1] 1.930891
```



```
## [1] "Skewness"  
## [1] 1.246533
```



```
## [1] "Num Outliers:"
## [1] 104
## [1] "Num. severe outliers:"
## [1] 17
## [1] "Shapiro test: "
##
## Shapiro-Wilk normality test
##
## data: data
## W = 0.93107, p-value < 2.2e-16
outliers_df[res.var,"birthrate"] <- 1
```

Data Quality Report

Now that all the variables have been slightly analyzed in terms of their meaning. Given that univariate outliers and NA values have already been computed and shown, only a ranking and a visual summary will be showcased to gather the main insights visually.

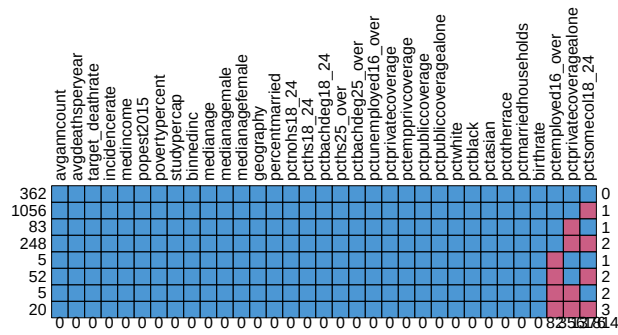
This quality report will be divided between variables and individuals of the dataset, and finally a multivariate outlier analysis will be performed.

Per variable

Per variable, count:

- Number of missing values.
- Number of errors (including inconsistencies).
- Number of outliers.
- Rank variables according the sum of missing values (and errors).

```
# Number of errors (including inconsistencies). TODO
md.pattern(train, rotate.names=T)
```



##	avganncount	avgdeathspereyear	target_deathrate	incidencerate	medianincome	
## 362	1	1	1	1	1	
## 1056	1	1	1	1	1	
## 83	1	1	1	1	1	
## 248	1	1	1	1	1	
## 5	1	1	1	1	1	
## 52	1	1	1	1	1	
## 5	1	1	1	1	1	
## 20	1	1	1	1	1	
##	0	0	0	0	0	
##	popest2015	povertypercent	studypercap	binnedinc	medianage	medianagemale
## 362	1	1	1	1	1	1
## 1056	1	1	1	1	1	1
## 83	1	1	1	1	1	1
## 248	1	1	1	1	1	1
## 5	1	1	1	1	1	1
## 52	1	1	1	1	1	1
## 5	1	1	1	1	1	1
## 20	1	1	1	1	1	1
##	0	0	0	0	0	0
##	medianagefemale	geography	percentmarried	pctnohs18_24	pcths18_24	
## 362	1	1	1	1	1	
## 1056	1	1	1	1	1	
## 83	1	1	1	1	1	
## 248	1	1	1	1	1	
## 5	1	1	1	1	1	
## 52	1	1	1	1	1	
## 5	1	1	1	1	1	
## 20	1	1	1	1	1	
##	0	0	0	0	0	
##	pctbachdeg18_24	pcths25_over	pctbachdeg25_over	pctunemployed16_over		
## 362	1	1	1	1		
## 1056	1	1	1	1		
## 83	1	1	1	1		
## 248	1	1	1	1		
## 5	1	1	1	1		
## 52	1	1	1	1		
## 5	1	1	1	1		
## 20	1	1	1	1		
##	0	0	0	0		

```

##      pctprivatecoverage pctempprivcoverage pctpubliccoverage
## 362                1                1                1
## 1056               1                1                1
## 83                1                1                1
## 248               1                1                1
## 5                 1                1                1
## 52               1                1                1
## 5                 1                1                1
## 20               1                1                1
##                  0                0                0
##      pctpubliccoveragealone pctwhite pctblack pctasian pctotherrace
## 362                1                1                1                1
## 1056               1                1                1                1
## 83                1                1                1                1
## 248               1                1                1                1
## 5                 1                1                1                1
## 52               1                1                1                1
## 5                 1                1                1                1
## 20               1                1                1                1
##                  0                0                0                0
##      pctmarriedhouseholds birthrate pctemployed16_over pctprivatecoveragealone
## 362                1                1                1                1
## 1056               1                1                1                1
## 83                1                1                1                0
## 248               1                1                1                0
## 5                 1                1                0                1
## 52               1                1                0                1
## 5                 1                1                0                0
## 20               1                1                0                0
##                  0                0                82                356
##      pctsomecol18_24
## 362                1    0
## 1056               0    1
## 83                1    1
## 248               0    2
## 5                 1    1
## 52               0    2
## 5                 1    2
## 20               0    3
##                  1376 1814

```

On one hand, looking at the pattern, we can notice that there are 362 variables with no null values, and that the vast majority of nulls (1056) are located on the pctsomecol18_24 variable with just 1 null column. Another remarkable trend is that up to 248 individuals have 2 null values in variables pctsomecol18_24 and pctprivatecoveragealone, while this other variable just has 83 pure nulls (Only this column). The last variable that contains a low amount of null values is pctemployed16_over, where 20 individuals have the last 3 mentioned variables containing nulls, and the rest of few possible NA values come from pattern permutations of these 3 variables. Therefore this dataset only has 3 possible rows containing NA, and only one of them is relevant in this aspect. Ranking:

Rank	Null values
1	pctsomecol18_24: 1375 NA
2	pctprivatecoveragealone: 365 NA
4	avganncount : 126 NA

Rank	Null values
4	incidencerate : 126 NA
3	pctemployed16_over: 82 NA

On the other hand, the top 5 ranking for variables with the most amount of mild and severe outliers is:

Rank	Mild Outliers	Severe Outliers
1	avganncount 333	avganncount 257
2	pctblack 323	studypercap 222
3	studypercap 307	popest2015 162
4	pctasian 305	avgdeathsperyear 138
5	pctotherrace 283	pctasian 107

Per individuals

Per individuals, count:

- Number of missing values
- Number of errors
- Number of outliers
- Identify individuals considered as multivariant outliers.

There are 362 obs without NA in any variable, 1144 obs with 1 variable with a NA value, 305 obs with 2 NA values and 20 obs with 3 NA values along the 33 variables of the dataset.

```
#Number of missing values
na_counts <- rowSums(is.na(train))
table(na_counts)
```

```
## na_counts
##    0    1    2    3
## 362 1144  305   20
```

Out of all observations, 836 do not exhibit any outliers across the variables, while 451 observations contain a single outlier in one variable. Interestingly, there are 2 observations identified as outliers in 10 variables and another 2 observations that are outliers in 11 variables.

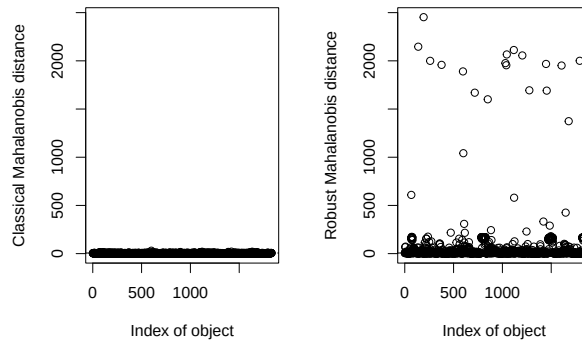
```
#Number of outliers per individual
outliers_counts <- rowSums(outliers_df)
table(outliers_counts)
```

```
## outliers_counts
##    0    1    2    3    4    5    6    7    8    9   10   11
## 836 451 163 115  95  71  52  18  19   7   2   2
```

Considering over 20 numerical variables that contain few correlated variables and considering a 1% significance level, 209 multivariate outliers are found in the dataset.

```
#Identify individuals considered as multivariant outliers.
train$avgdeathsperyear <- as.numeric(train$avgdeathsperyear)
train$medincome <- as.numeric(train$medincome)
train$popest2015 <- as.numeric(train$popest2015)

#Compute multivariant outliers and find their ids (24)
subset_num_vars <- c(1,2,3,7,10:12,14:16, 18:20, 22,23, 25:33)
res_mout <- Moutlier(train[,subset_num_vars], quantile=0.97, plot=T)
```



```
mout_ids <- which((res_mout$md > res_mout$cutoff) & (res_mout$rd > res_mout$cutoff))

#Reset graphical parameters and plot the outliers
dev.off()
```

```
## null device
##          1
```

```
plot(x=res_mout$md, y=res_mout$rd, main="Classic Mahalanobis D vs Robust Mahalanobis D (M. Outliers)",
abline(v=res_mout$cutoff, col="red")
abline(h=res_mout$cutoff, col="red")
```

We can clearly see that for the number of outliers variable, the mean is around 1.39, and the third quantile around 2.0, however, we have a high standard deviation of 1.935 pointing out at the presence of outliers in the number of outliers variable, suggesting there are several individuals with more than 2 outlier variables. This variable contains somewhat high correlation with pcths25_over, pchwhite, pctasian and other weak correlations.

The number of missing values doesn't make much sense to be analyzed after removing 2 of the main problematic columns which contained several null values.

Finally, looking at the multivariate factor profiling, we see that there is a global association between "mout" and "binnedinc", and we can see that there is a huge global association with quantitative variables pctotherrace and n_uout (Logically, the number of univariate outliers contributes to a possible multivariate outlier individual), followed by medianage looking at their eta squared values. Finally, we see looking at the mean values of the quantitative variables that in the case of an individual that is not a multivariate outlier, almost every single variable mean value is equal or close to the overall mean. However in the case of multivariate outliers we see huge discrepancies in the mean of pctasian, pctotherrace, n_out, popest2015, avgdeathspereyear and avganncount, which are way higher than the overall mean value. This may point out that the asian-otherrace populations and those with higher population or more average deaths per year, are more prone to be multivariate outliers.

```
anal_df <- train
#Binary factor if its a multivariate outlier
anal_df$mout <- 0
anal_df[mout_ids, "mout"] <- 1
anal_df$mout <- factor(anal_df$mout, labels=c("Not M.O", "Yes M.O"))
#Number of univariate outlier variables for a given observation
anal_df$n_uout <- rowSums(outliers_df)
#Number of missing values in variables for a given observation
anal_df$n_miss <- rowSums(is.na(train))
```



```

## binnedinc=(61494.5, 125635] 20.430108 16.888889 10.158383 0.0008458263
## binnedinc=[22640, 34218.1] 19.587629 16.888889 10.595303 0.0020447042
## binnedinc=(48021.6, 51046.4] 7.692308 6.666667 10.649918 0.0320467887
## binnedinc=(51046.4, 54545.6] 6.250000 4.888889 9.612234 0.0064710699
##
## v.test
## binnedinc=(61494.5, 125635] 3.337346
## binnedinc=[22640, 34218.1] 3.083661
## binnedinc=(48021.6, 51046.4] -2.143827
## binnedinc=(51046.4, 54545.6] -2.722905
##
##
## Link between the cluster variable and the quantitative variables
## =====
##
## Eta2 P-value
## n_uout 0.155323819 4.286531e-69
## pctotherrace 0.099628021 1.225624e-43
## medianage 0.062554722 1.635718e-27
## pctwhite 0.059116765 4.782414e-26
## pctasian 0.045841171 1.988142e-20
## pcths25_over 0.043595539 1.748772e-19
## popest2015 0.034525185 1.098770e-15
## avganncount 0.032527270 7.489993e-15
## avgdeathsperyear 0.030126599 7.496950e-14
## pctempprivcoverage 0.019328031 2.316662e-09
## percentmarried 0.019175132 2.681751e-09
## incidencerate 0.016324389 4.112895e-08
## pctnohs18_24 0.015285419 1.113888e-07
## pctprivatecoverage 0.013413434 6.724710e-07
## povertypercent 0.013283435 7.620289e-07
## target_deathrate 0.011450565 4.455514e-06
## medianagefemale 0.010053541 1.720632e-05
## medianagemale 0.009396851 3.253972e-05
## birthrate 0.009256862 3.728142e-05
## pctbachdeg25_over 0.009091283 4.379394e-05
## pctbachdeg18_24 0.008074303 1.180369e-04
## pctmarriedhouseholds 0.007426363 2.226259e-04
## pctunemployed16_over 0.005856003 1.048909e-03
## pctemployed16_over 0.004120889 7.241760e-03
## pctprivatecoveragealone 0.004498010 9.981131e-03
## pctsomecol18_24 0.014549394 1.001640e-02
## medincome 0.003262597 1.450668e-02
##
## Description of each cluster by quantitative variables
## =====
## $`Not M.O`
##
## v.test Mean in category Overall mean sd in category
## pctwhite 10.401138 85.306077 8.384699e+01 1.435103e+01
## pcths25_over 8.931956 35.284620 3.473255e+01 6.715734e+00
## pctempprivcoverage 5.947293 41.789601 4.129388e+01 8.979987e+00
## percentmarried 5.923723 52.250125 5.190087e+01 6.086924e+00
## incidencerate 5.465678 451.661576 4.489919e+02 4.730818e+01
## pctsomecol18_24 5.372873 40.996456 4.047846e+01 1.048402e+01
## pctprivatecoverage 4.954451 64.935554 6.447460e+01 9.772785e+00
## target_deathrate 4.577612 179.938045 1.788197e+02 2.528725e+01

```

```

## medianagefemale      4.289287      42.364259 4.216931e+01 4.743624e+00
## medianagemale        4.146835      39.773848 3.958564e+01 4.733155e+00
## pctmarriedhouseholds 3.686495      51.610338 5.140232e+01 5.883973e+00
## pctprivatecoveragealone 2.893764      48.902943 4.864807e+01 9.470267e+00
## pctemployed16_over    2.724793      54.407547 5.421172e+01 7.621998e+00
## medincome            -2.443472      47014.155044 4.727794e+04 1.115646e+04
## pctunemployed16_over -3.273604      7.761955 7.860950e+00 3.096049e+00
## pctbachdeg18_24      -3.843953      6.064321 6.216221e+00 3.813470e+00
## pctbachdeg25_over     -4.078854      13.102366 1.329754e+01 5.103788e+00
## birthrate            -4.115830      5.527922 5.597439e+00 1.694128e+00
## povertypercent        -4.930384      16.515255 1.679175e+01 5.851570e+00
## pctnohs18_24         -5.288886      17.921357 1.828777e+01 7.111371e+00
## avgdeathsperyear     -7.425071      154.480697 1.916237e+02 2.701091e+02
## avganncount          -7.715238      516.055275 6.232245e+02 8.349126e+02
## popest2015           -7.948653      80663.803861 1.068412e+05 1.580656e+05
## pctasian             -9.159113      1.066889 1.274301e+00 1.550148e+00
## medianage            -10.699306      41.051868 4.524970e+01 4.701650e+00
## pctotherrace         -13.502566      1.569335 2.003134e+00 2.271112e+00
## n_uout               -16.859496      1.108344 1.393774e+00 1.642968e+00
##
## Overall sd      p.value
## pctwhite        1.603270e+01 2.449885e-25
## pcths25_over    7.064060e+00 4.185422e-19
## pctempprivcoverage 9.526266e+00 2.726128e-09
## percentmarried  6.738289e+00 3.147337e-09
## incidencerate   5.582455e+01 4.611418e-08
## pctsomocol18_24 1.101857e+01 7.749189e-08
## pctprivatecoverage 1.063324e+01 7.253484e-07
## target_deathrate 2.792279e+01 4.703134e-06
## medianagefemale  5.194588e+00 1.792479e-05
## medianagemale    5.187250e+00 3.371034e-05
## pctmarriedhouseholds 6.448935e+00 2.273640e-04
## pctprivatecoveragealone 1.006636e+01 3.806547e-03
## pctemployed16_over 8.213804e+00 6.434182e-03
## medincome        1.233832e+04 1.454672e-02
## pctunemployed16_over 3.456167e+00 1.061851e-03
## pctbachdeg18_24  4.516322e+00 1.210682e-04
## pctbachdeg25_over 5.468847e+00 4.525832e-05
## birthrate        1.930363e+00 3.857885e-05
## povertypercent    6.409410e+00 8.206812e-07
## pctnohs18_24     7.917877e+00 1.230634e-07
## avgdeathsperyear 5.717202e+02 1.127197e-13
## avganncount       1.587552e+03 1.207562e-14
## popest2015        3.763912e+05 1.885499e-15
## pctasian          2.588138e+00 5.232582e-20
## medianage         4.484112e+01 1.025437e-26
## pctotherrace      3.671806e+00 1.510237e-41
## n_uout            1.934917e+00 8.934843e-64
##
## $`Yes M.O`
##
## v.test Mean in category Overall mean sd in category
## n_uout      16.859496      3.431111e+00 1.393774e+00 2.543123e+00
## pctotherrace 13.502566      5.099498e+00 2.003134e+00 7.871966e+00
## medianage    10.699306      7.521289e+01 4.524970e+01 1.232132e+02
## pctasian     9.159113      2.754761e+00 1.274301e+00 5.904230e+00

```

## popest2015	7.948653	2.936894e+05	1.068412e+05	9.668209e+05
## avganncount	7.715238	1.388175e+03	6.232245e+02	3.855787e+03
## avgdeathsperyear	7.425071	4.567422e+02	1.916237e+02	1.434938e+03
## pctnohs18_24	5.288886	2.090311e+01	1.828777e+01	1.189171e+01
## povertypercent	4.930384	1.876533e+01	1.679175e+01	9.244450e+00
## birthrate	4.115830	6.093634e+00	5.597439e+00	3.091473e+00
## pctbachdeg25_over	4.078854	1.469067e+01	1.329754e+01	7.432688e+00
## pctbachdeg18_24	3.843953	7.300444e+00	6.216221e+00	7.800373e+00
## pctunemployed16_over	3.273604	8.567556e+00	7.860950e+00	5.312055e+00
## medincome	2.443472	4.916081e+04	4.727794e+04	1.861157e+04
## pctemployed16_over	-2.767633	5.279198e+01	5.421172e+01	1.153767e+01
## pctprivatecoveragealone	-2.844516	4.685978e+01	4.864807e+01	1.339354e+01
## pctmarriedhouseholds	-3.686495	4.991756e+01	5.140232e+01	9.423835e+00
## medianagemale	-4.146835	3.824222e+01	3.958564e+01	7.550120e+00
## medianagefemale	-4.289287	4.077778e+01	4.216931e+01	7.534321e+00
## target_deathrate	-4.577612	1.708369e+02	1.788197e+02	4.132807e+01
## pctprivatecoverage	-4.954451	6.118444e+01	6.447460e+01	1.503502e+01
## pctsomecol18_24	-4.955522	3.706833e+01	4.047846e+01	1.355299e+01
## incidencerate	-5.465678	4.299362e+02	4.489919e+02	9.471860e+01
## percentmarried	-5.923723	4.940800e+01	5.190087e+01	9.896851e+00
## pctempprivcoverage	-5.947293	3.775556e+01	4.129388e+01	1.219163e+01
## pcths25_over	-8.931956	3.079200e+01	3.473255e+01	8.152171e+00
## pctwhite	-10.401138	7.343237e+01	8.384699e+01	2.231803e+01
##	Overall sd	p.value		
## n_uout	1.934917e+00	8.934843e-64		
## pctotherrace	3.671806e+00	1.510237e-41		
## medianage	4.484112e+01	1.025437e-26		
## pctasian	2.588138e+00	5.232582e-20		
## popest2015	3.763912e+05	1.885499e-15		
## avganncount	1.587552e+03	1.207562e-14		
## avgdeathsperyear	5.717202e+02	1.127197e-13		
## pctnohs18_24	7.917877e+00	1.230634e-07		
## povertypercent	6.409410e+00	8.206812e-07		
## birthrate	1.930363e+00	3.857885e-05		
## pctbachdeg25_over	5.468847e+00	4.525832e-05		
## pctbachdeg18_24	4.516322e+00	1.210682e-04		
## pctunemployed16_over	3.456167e+00	1.061851e-03		
## medincome	1.233832e+04	1.454672e-02		
## pctemployed16_over	8.213804e+00	5.646501e-03		
## pctprivatecoveragealone	1.006636e+01	4.447890e-03		
## pctmarriedhouseholds	6.448935e+00	2.273640e-04		
## medianagemale	5.187250e+00	3.371034e-05		
## medianagefemale	5.194588e+00	1.792479e-05		
## target_deathrate	2.792279e+01	4.703134e-06		
## pctprivatecoverage	1.063324e+01	7.253484e-07		
## pctsomecol18_24	1.101857e+01	7.213639e-07		
## incidencerate	5.582455e+01	4.611418e-08		
## percentmarried	6.738289e+00	3.147337e-09		
## pctempprivcoverage	9.526266e+00	2.726128e-09		
## pcths25_over	7.064060e+00	4.185422e-19		
## pctwhite	1.603270e+01	2.449885e-25		

Imputation

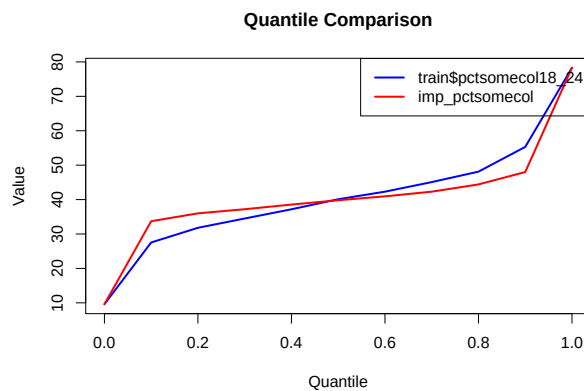
pctsomecol18_24

Given the huge amount of null values, the impute PCA has not been able to replicate the original distribution of this messy variable.

```
num_vars <- train[sapply(train, is.numeric)]
miss_ids <- which(is.na(train$pctsomecol18_24))
res_imp <- imputePCA(num_vars)
imp_pctsomecol <- res_imp$completeObs[, "pctsomecol18_24"]

# Calculate the quantiles for validation
quantiles_train <- quantile(train$pctsomecol18_24, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_pctsomecol, probs = seq(0, 1, 0.1), na.rm = FALSE)

plot(seq(0, 1, 0.1), quantiles_train, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c("train$pctsomecol18_24", "imp_pctsomecol"),
      col = c("blue", "red"), lwd = 2)
```



```
#We won't impute the values!
train[miss_ids, "pctsomecol18_24"] <- imp_pctsomecol[miss_ids]
```

pctemployed16_over

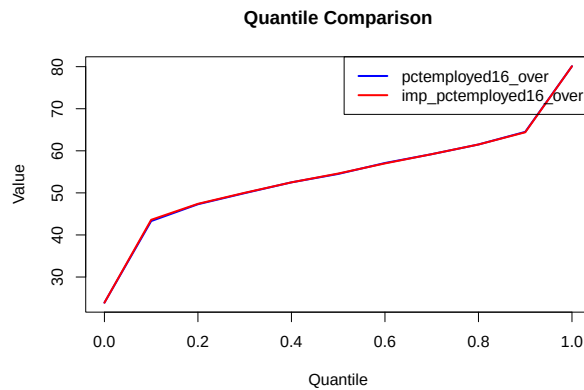
pctemployed16_over only contains 70 null values and the imputePCA function has been able of almost perfectly preserve the original distribution of the variable. Therefore, it is indeed worth to perform the imputation on the train data.

```
num_vars <- train[sapply(train, is.numeric)]
miss_ids <- which(is.na(train$pctemployed16_over))
res_imp <- imputePCA(num_vars)
imp_pctemployed16_over <- res_imp$completeObs[, "pctemployed16_over"]

# Calculate the quantiles for validation
quantiles_train <- quantile(train$pctemployed16_over, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_pctemployed16_over, probs = seq(0, 1, 0.1), na.rm = FALSE)

plot(seq(0, 1, 0.1), quantiles_train, type = "l", col = "blue", lwd = 2,
      xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
```

```
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c("pctemployed16_over", "imp_pctemployed16_over"),
      col = c("blue", "red"), lwd = 2)
```



```
train[miss_ids, "pctemployed16_over"] <- imp_pctemployed16_over[miss_ids]
```

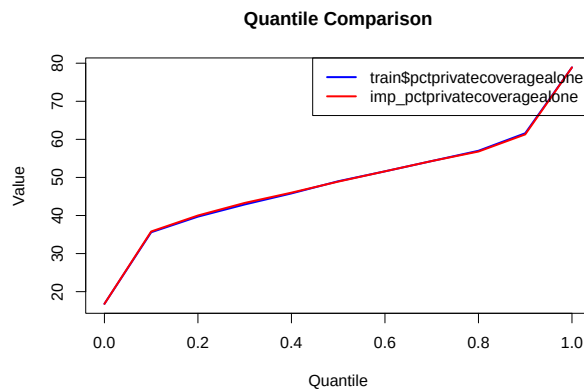
pctprivatecoveragealone

Similarly, the variable pctprivatecoveragealone is better informed despite a quite high number of null values (253) and the imputePCA has been successful to preserve the original distribution of the variable. Therefore, it is indeed worth to perform the imputation on the train data.

```
num_vars <- train[sapply(train, is.numeric)]
miss_ids <- which(is.na(train$pctprivatecoveragealone))
res_imp <- imputePCA(num_vars)
imp_pctprivatecoveragealone <- res_imp$completeObs[, "pctprivatecoveragealone"]

# Calculate the quantiles for validation
quantiles_train <- quantile(train$pctprivatecoveragealone, probs = seq(0, 1, 0.1), na.rm = TRUE)
quantiles_imp <- quantile(imp_pctprivatecoveragealone, probs = seq(0, 1, 0.1), na.rm = FALSE)

plot(seq(0, 1, 0.1), quantiles_train, type = "l", col = "blue", lwd = 2,
     xlab = "Quantile", ylab = "Value", main = "Quantile Comparison")
lines(seq(0, 1, 0.1), quantiles_imp, col = "red", lwd = 2)
legend("topright", legend = c("train$pctprivatecoveragealone", "imp_pctprivatecoveragealone"),
      col = c("blue", "red"), lwd = 2)
```



```
train[miss_ids, "pctprivatecoveragealone"] <- imp_pctprivatecoveragealone[miss_ids]
```

Profiling & Relationships

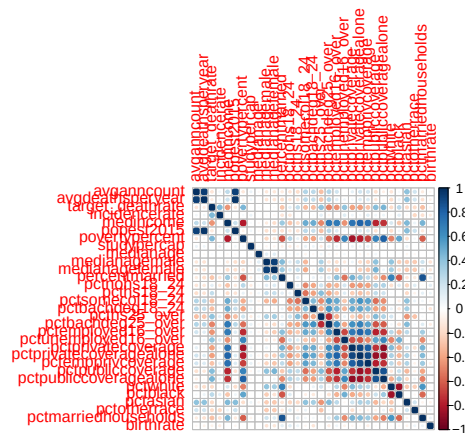
After imputation and removal of null values, we can proceed with profiling the target variable and assess if any variable contains serial correlation.

TODO: explicar correlacions... (Cuidado, no cal explicar les correlacions de target_deathrate, ja estan just al apartat d'abaix!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!)

```
names(train)
```

```
## [1] "avganncount"          "avgdeathsperyear"
## [3] "target_deathrate"    "incidencerate"
## [5] "medincome"           "popest2015"
## [7] "povertypercent"      "studypercap"
## [9] "binnedinc"           "medianage"
## [11] "medianagemale"       "medianagefemale"
## [13] "geography"           "percentmarried"
## [15] "pctnohs18_24"        "pcths18_24"
## [17] "pctsomecol18_24"     "pctbachdeg18_24"
## [19] "pcths25_over"        "pctbachdeg25_over"
## [21] "pctemployed16_over"  "pctunemployed16_over"
## [23] "pctprivatecoverage"  "pctprivatecoveragealone"
## [25] "pctempprivcoverage"  "pctpubliccoverage"
## [27] "pctpubliccoveragealone" "pctwhite"
## [29] "pctblack"            "pctasian"
## [31] "pctotherrace"        "pctmarriedhouseholds"
## [33] "birthrate"
```

```
num_vars <- train[sapply(train, is.numeric)]
cat_vars <- train[!sapply(train, is.numeric)]
corrplot(cor(num_vars))
```



For some unexpected reason the geography categorical variable causes some unexpected problems to arise, therefore we won't use it at all. If you take a closer look at the profiling of this target variable you will realize that there is a high linkage between the target and the qualitative variable binnedinc performing a oneway anova test. If you take a look at the mean values for different values of binnedinc, you will see that those incomes which are below 40K have a higher deathrate (The lower the binned income the higher the deathrate). Conversely, the higher the binned income the lower the deathrate. Finally, taking a glance at the quantitative variables, we see that there is a medium-high positive correlation with pctpubliccoveragealone,

povertypercent, pctpubliccoverage, pcths25_over and pctunemployed16_over. Looking at the inverse, we see a medium-high inverse correlation with variables medincome, pctbachdeg25_over and pctprivatecoverage and pctmarriedhouseholds.

```
subset_num <- c(1,2,3:7,10:12,14:16, 18:20, 22,23, 25:33)
subset_cat <- c(9)
subset_train <- c(subset_num, subset_cat)

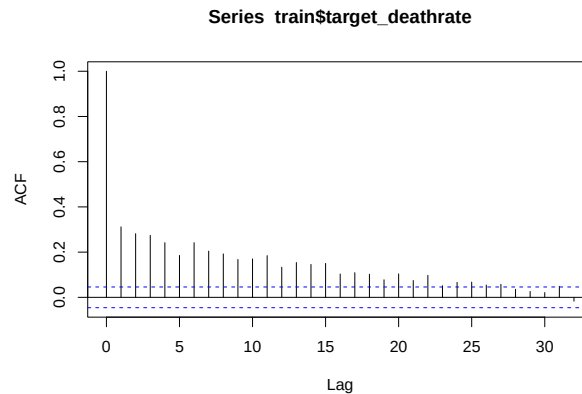
res_anal <- condres(train[, subset_train], num.var=3, proba=0.01)
res_anal
```

```
##
## Link between the variable and the continuous variables (R-square)
## =====
##               correlation      p.value
## pctpubliccoveragealone  0.46629578  1.687857e-99
## povertypercent          0.45008465  5.144264e-92
## incidencerate           0.44954847  8.953861e-92
## pctpubliccoverage       0.42436751  6.056303e-81
## pcths25_over            0.40690172  5.910723e-74
## pctunemployed16_over    0.39780681  1.777035e-70
## pctblack                0.26875452  1.156196e-31
## pcths18_24              0.24243487  6.687515e-26
## pctnohs18_24            0.11480060  8.424149e-07
## birthrate               -0.06756043  3.824884e-03
## avgdeathsperyear        -0.09167650  8.549496e-05
## popest2015              -0.11902232  3.261480e-07
## avganncount             -0.13886454  2.417857e-09
## pctwhite                -0.18152553  4.983753e-15
## pctotherrace            -0.18153593  4.965708e-15
## pctasian                -0.19436412  4.806194e-17
## percentmarried          -0.27253733  1.511951e-32
## pctempprivcoverage      -0.28877690  1.677032e-36
## pctmarriedhouseholds    -0.29440643  6.182702e-38
## pctbachdeg18_24         -0.30162255  8.054861e-40
## pctprivatecoverage      -0.40354149  1.173008e-72
## medincome               -0.44368280  3.606607e-89
## pctbachdeg25_over       -0.49777156  3.106333e-115
##
## Link between the variable and the categorical variable (1-way anova)
## =====
##               R2      p.value
## binnedinc 0.2217324 7.669702e-93
##
## Link between variable and the categories of the categorical variables
## =====
##               Estimate      p.value
## binnedinc=[22640, 34218.1] 25.763711 9.853130e-45
## binnedinc=(34218.1, 37413.8] 13.511997 1.196419e-11
## binnedinc=(37413.8, 40362.7]  9.454553 3.683926e-06
## binnedinc=(48021.6, 51046.4] -7.952886 3.574891e-05
## binnedinc=(51046.4, 54545.6] -9.460955 3.130385e-06
## binnedinc=(54545.6, 61494.5] -13.093413 1.157068e-11
## binnedinc=(61494.5, 125635] -20.458908 2.459391e-26
```

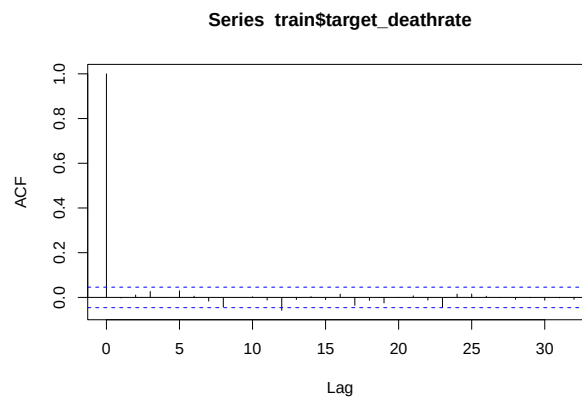
Serial correlations

It has been found previously that the target_variable doesn't follow a normal distribution at all, now its clear looking at the auto correlation function plot below that this variable contains serial correlation. To address this issue we will

```
library(stats)
acf(train$target_deathrate)
```



```
ids <- sample(1:nrow(train), nrow(train))
train <- train[ids, ]
acf(train$target_deathrate)
```



Modeling (TRAIN DATA)

#Include Code

Model exploration

Multicollinearity study

#Include Code

Model transformations

#Include Code

Model validation

In this section we will assess how well the model performs on the training data (train split) to ensure its robustness and capabilities, so we can assess under/over fitting.

We will analyze the residuals to ensure that model assumptions are met and use R-squared as the main performance metric will the model will be computed by means of Ordinary least squares.

#Include Code

Model evaluation

Finally we will evaluate how well the model performs with unseen data (train split) to assess its forecasting capabilities, using the same R-squared performance metric.

#Include Code